

Integration of an SiPM-Based Scintillation Detector into the Phase-2 CMS Drift Tube Readout Chain Using the IpGBT ASIC

Master's thesis in physics

Nils Jonathan Esper

Supervisor: Univ.-Prof. Dr. rer. nat. Alexander Schmidt

Second reviewer: Apl. Prof. Dr. rer. nat. Oliver Pooth



Physics Institute III A, RWTH Aachen University

Aachen, 19th December 2025

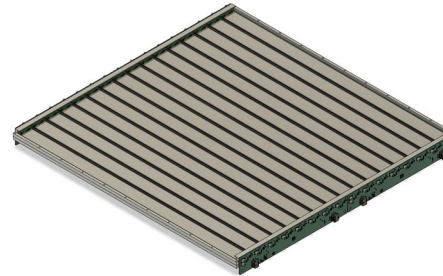
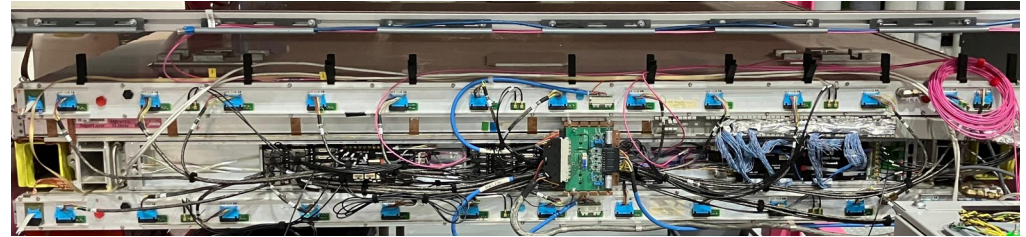
Introduction

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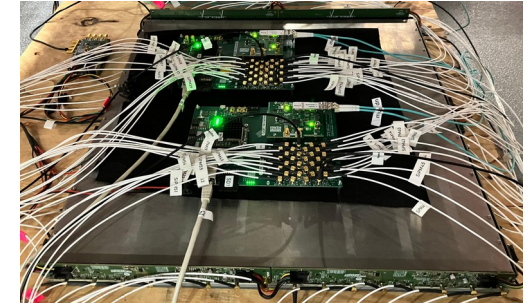
Two different muon detectors available:

- CMS Drift Tube (DT) chamber
 - Operated by our institute at cosmic test stand
- Scintillator strip detector with SiPM readout
 - Developed by our group over past theses

DT chamber at test stand



Scintillator rendering



Scintillator with electronics

Idea for this thesis:

- **Integrate scintillator into the CMS DT readout chain at test stand**
 - **Demonstrate synchronization of both detectors with atmospheric muon data**

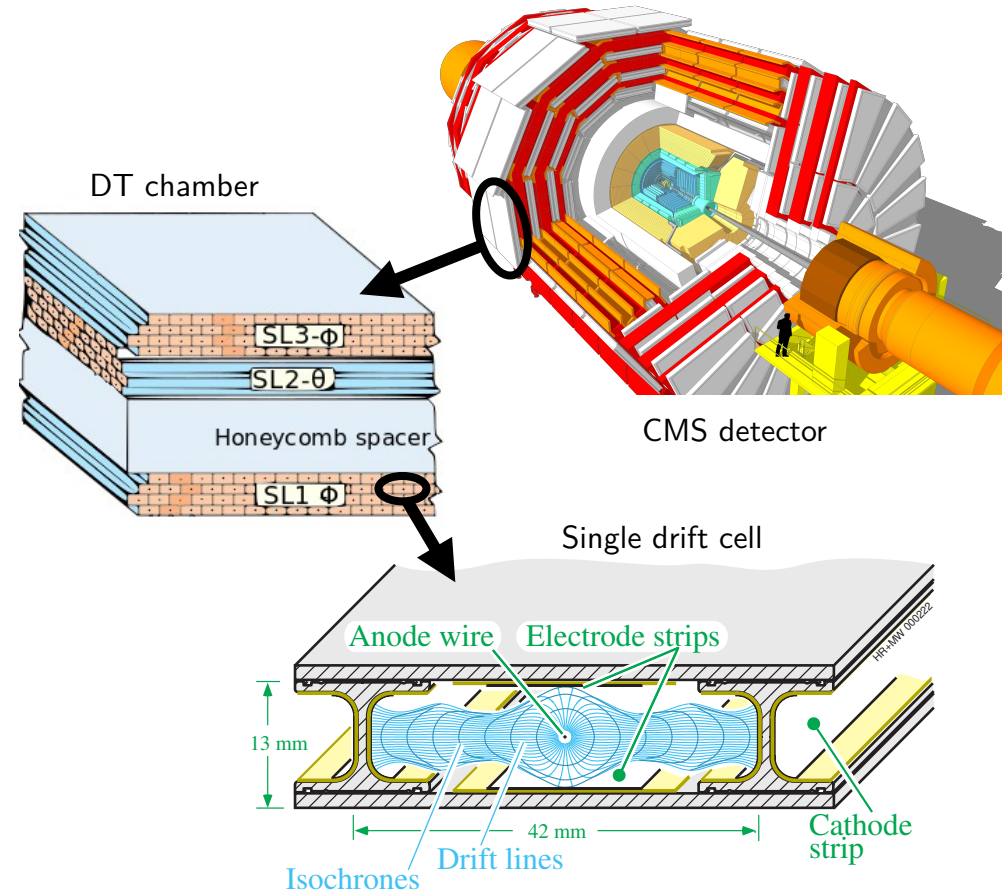
DT tasks: Recommission DT chamber with Phase-2 electronics & implement triggerless DT track reconstruction

Scintillator tasks: Develop new scintillator readout electronics & firmware

CMS Drift Tube chamber

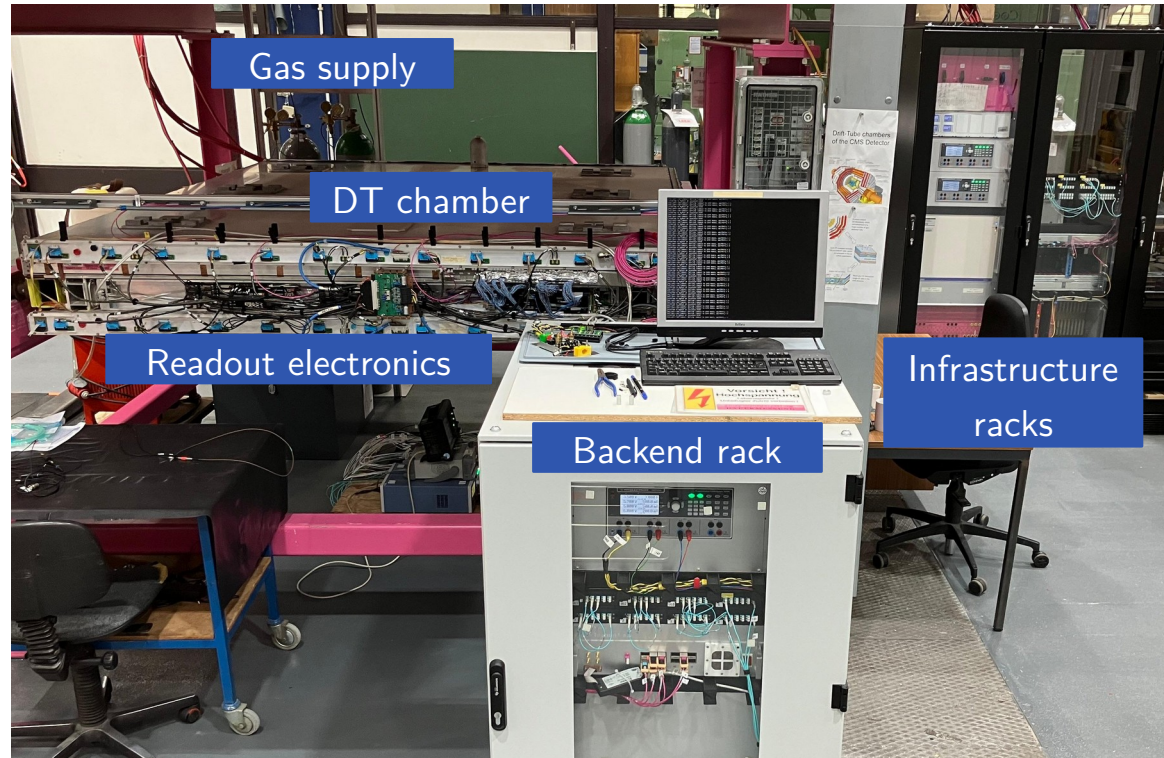
CMS DT system

- Compact Muon Solenoid (CMS) detector:
 - Multipurpose particle detector at Large Hadron Collider (LHC) at CERN
- Drift Tube (DT) chambers:
 - Gaseous detector in CMS barrel region $|\eta| < 1.2$
 - **3 superlayers** with **4 layers** of $O(50)$ **drift cells** each
 - SL 2 rotated relative to SLs 1, 3
- Drift cell:
 - Cross section: **42 x 13 mm²**, Length: $O(2\text{ m})$
 - Gas mixture: 85% Ar, 15% CO₂
 - Ionization electrons drift towards anode wire with $v_{\text{drift}} = 54.5\text{ }\mu\text{m/ns}$
 - Max. drift time of $t_{\text{drift,max}} = 385\text{ ns}$



Cosmic test stand in Aachen

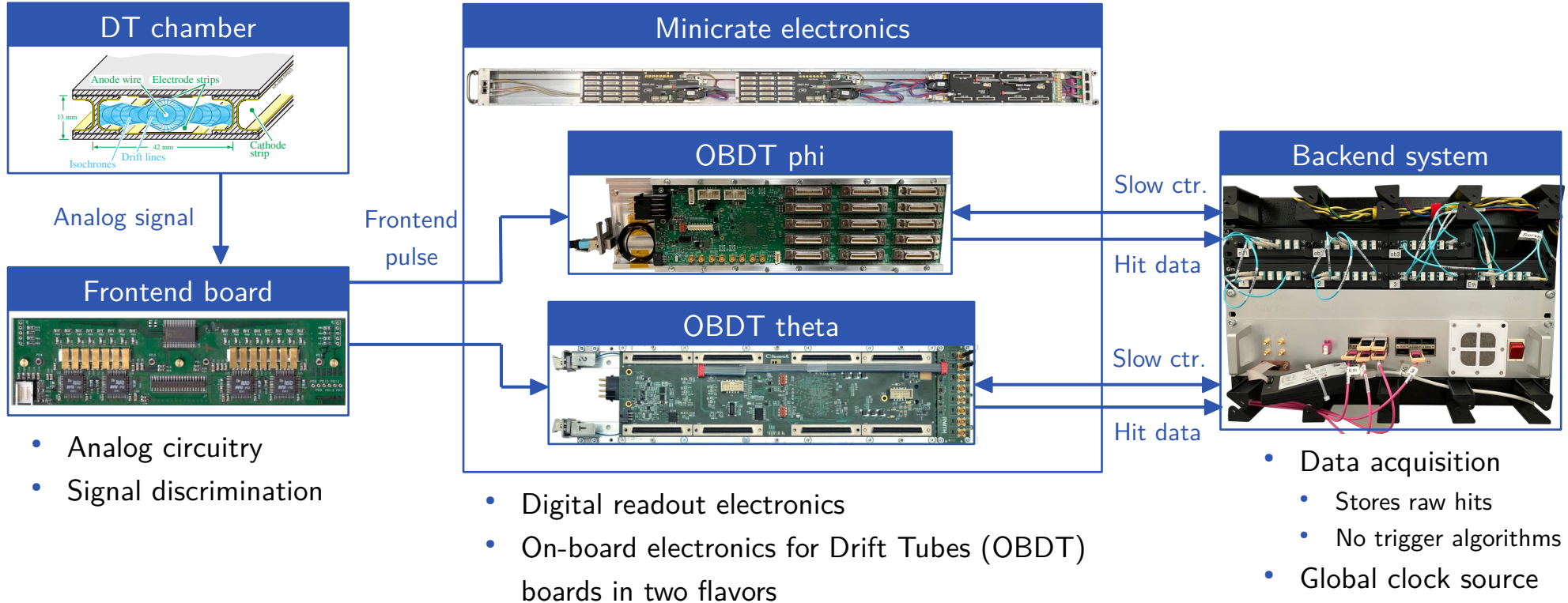
- Our institute operates cosmic test stand in experimental hall
 - Remainder of DT chamber production
- Features fully functional CMS DT chamber
 - Dimensions: $2.5 \times 2.2 \times 0.4 \text{ m}^3$



Setup at cosmic test stand

DT readout electronics

- ▶ Resembles CMS DT Phase-2 readout chain (except for backend system)



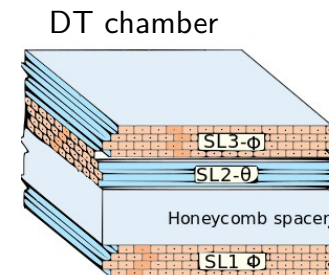
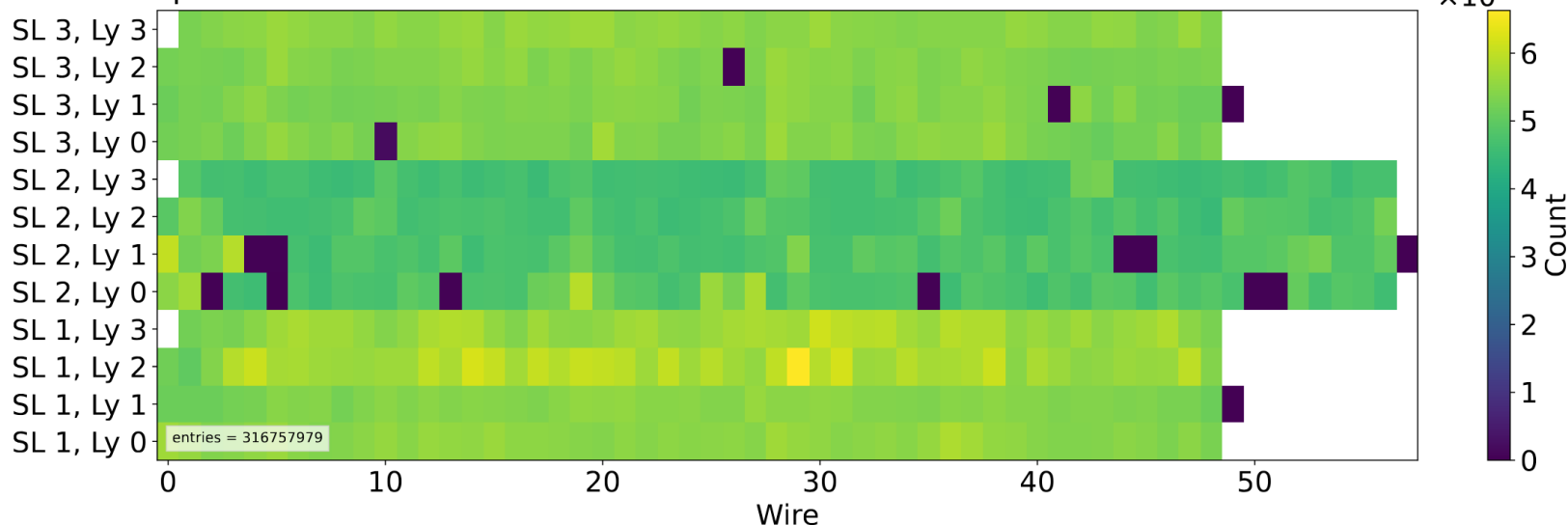
Recorded Drift Tube chamber data

DT hit data

- Recorded large dataset of atmospheric muons with DT chamber:

Duration	DT hits
5h 27min	316,757,979

Wire occupancies:



- Observe uniform wire rates: **28.0 Hz** (SLs 1, 3) & **24.5 Hz** (SL 2)
 - Rate differences because of different wire lengths (chamber size: 2.5 x 2.2 m²)
- 13 dead cells** & 3 unconnected cells in the chamber

Atmospheric muon simulation:
14.3 Hz (SLs 1, 3) & 12.3 Hz (SL 2)

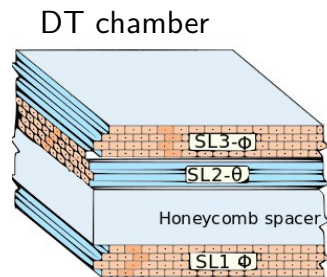
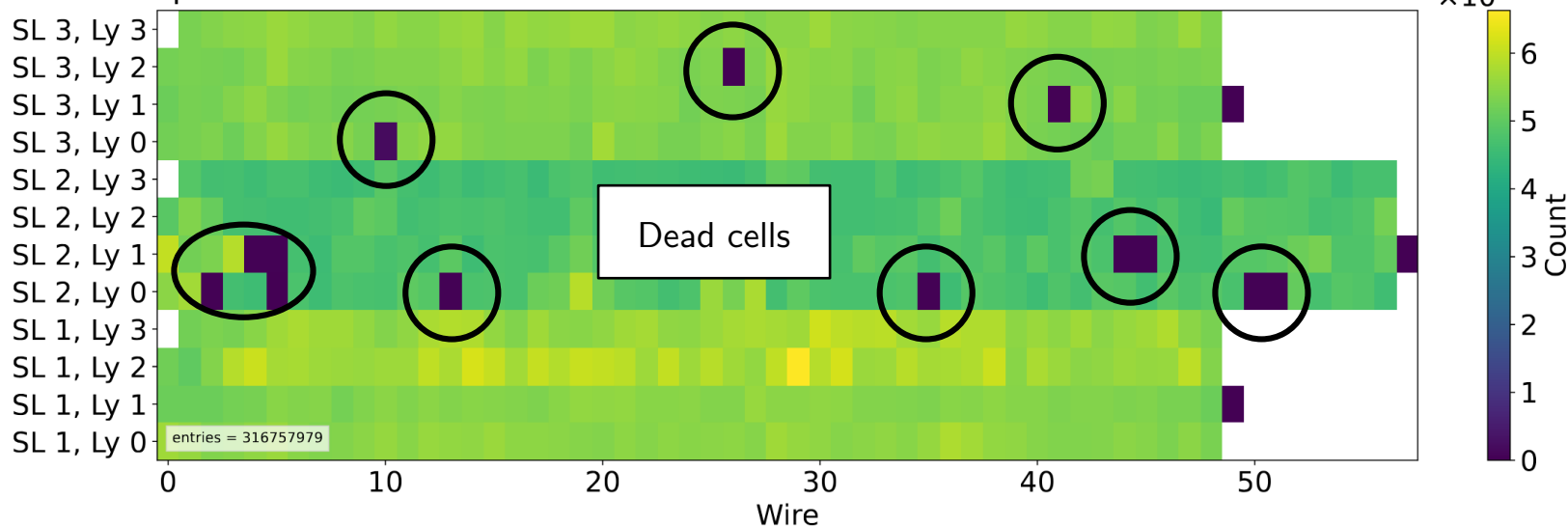
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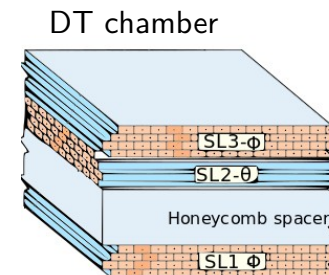
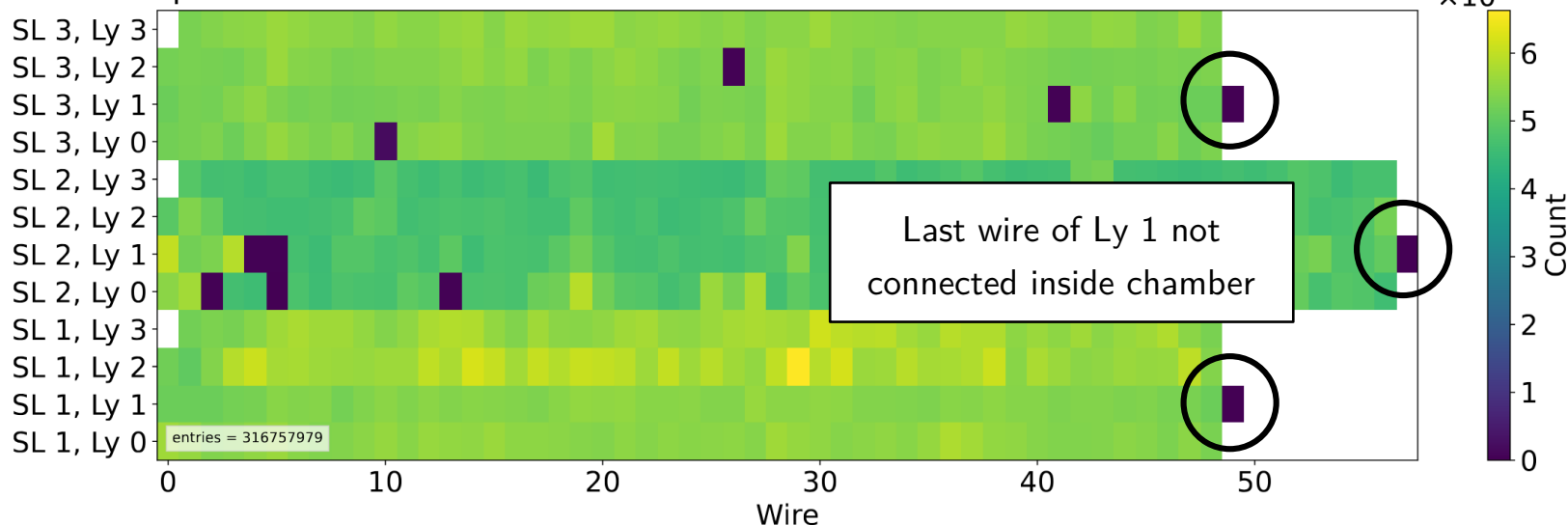
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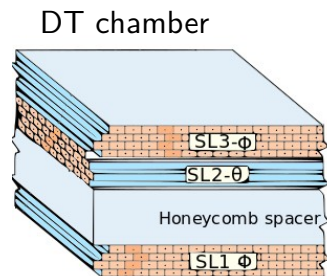
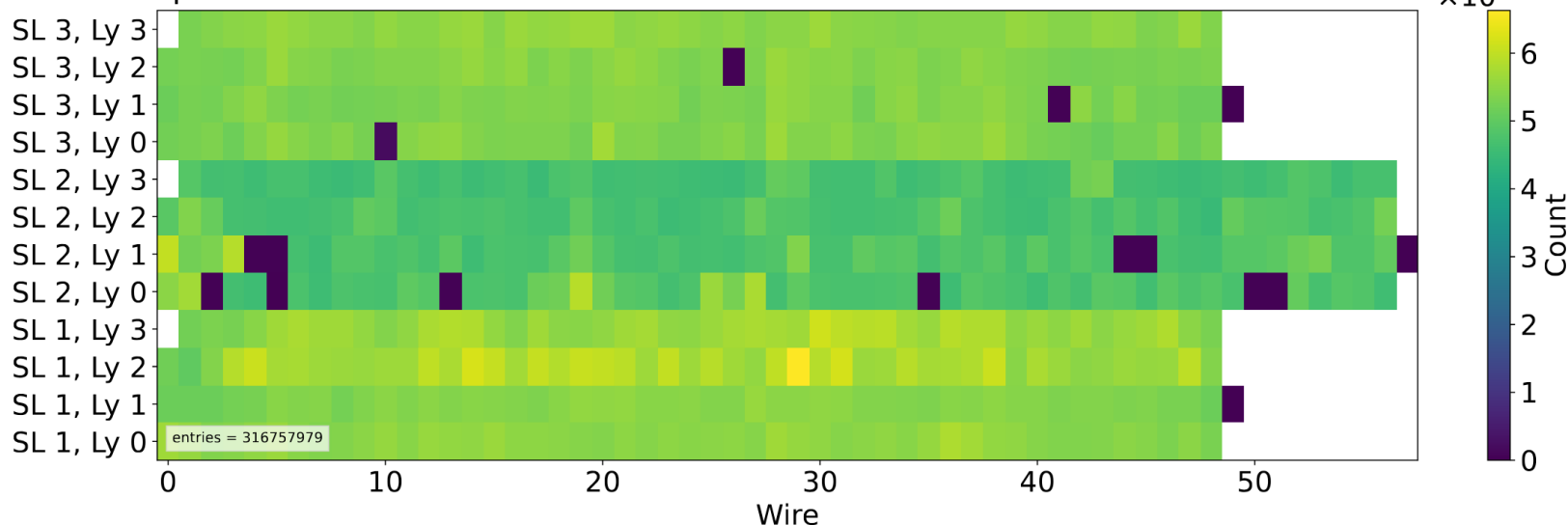
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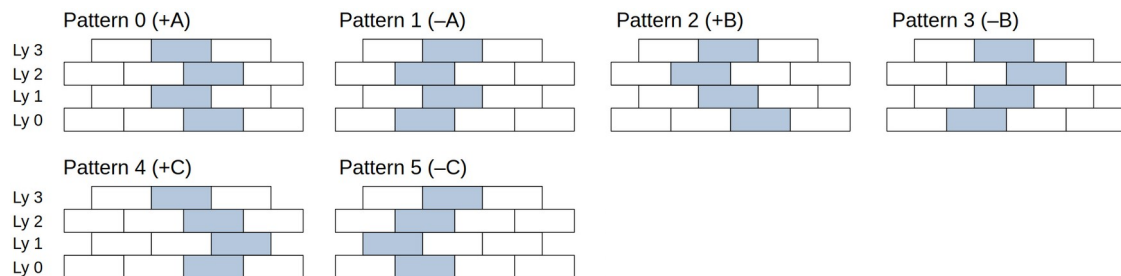
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DT superlayer patterns

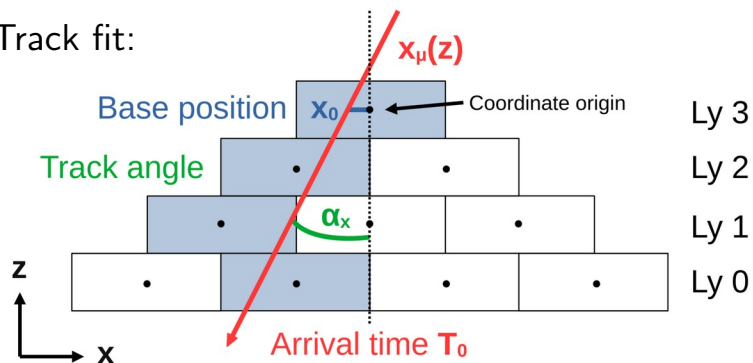
Triggerless track reconstruction:

- Treat all cell hits as potential track hits
- Superlayer pattern:
 - Exclude hits that could not be created simultaneously: $\Delta T_{\max} \leq t_{\text{drift},\max}$
 - Pattern should have **track-like shape**
 - Require 4 hits in pattern
- Superlayer pattern fit:
 - **Fit local track to pattern hits**
 - Relate hit timestamps to linear track (with simplifying assumptions)
 - **Local track parameters:** T_0 , x_0 , $\tan \alpha$
 - Apply cuts to reduce noise patterns
 - Goodness-of-fit variable $\chi^2/N_{\text{df}} < 10$

Superlayer pattern geometry:

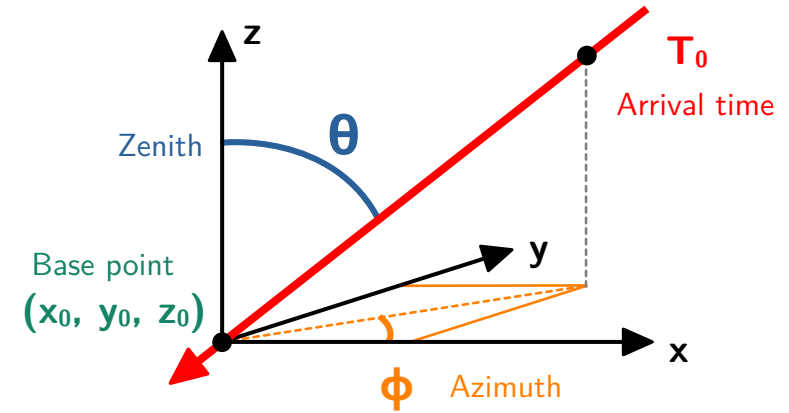


Track fit:



DT global track building

- Temporal matching of pattern fits between superlayers:
 - Select patterns close in time: $\Delta T_{0,SL} \leq 20 \text{ ns}$
- Translation of track projections into global frame:
 - **Global track parameters:** T_0 , (x_0, y_0, z_0) , ϕ , Θ



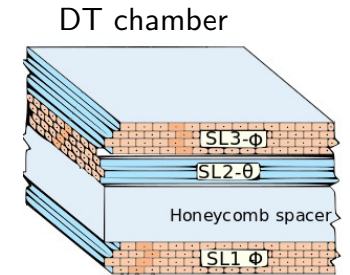
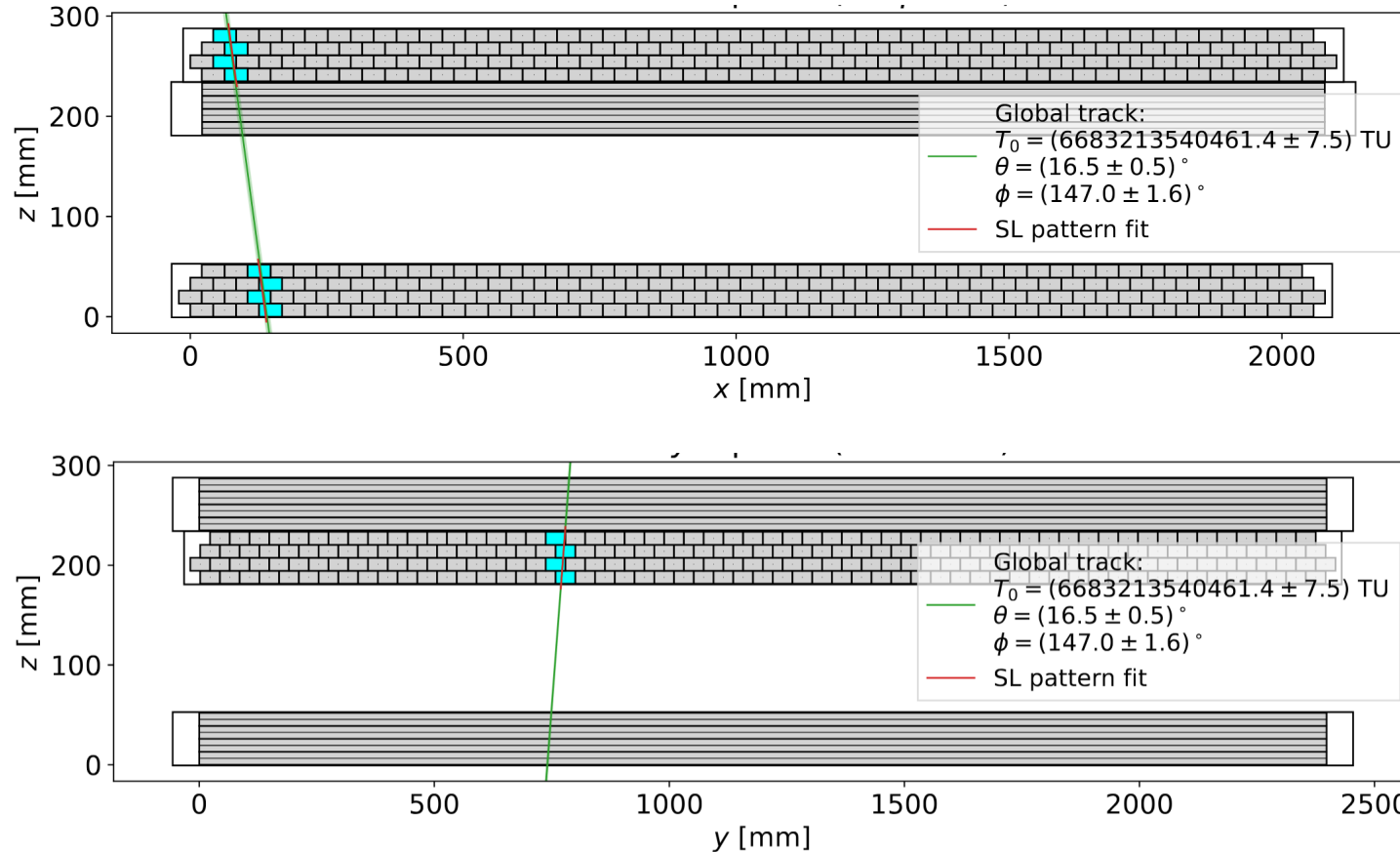
- ▶ Reconstructed global muon rate: **85 Hz**
- ▶ Global track parameters (mostly) behave as expected

Atmospheric muon simulation: 196 Hz

- ▶ Lower rate of reconstructed muons: Possibly because of assumptions in reconstruction

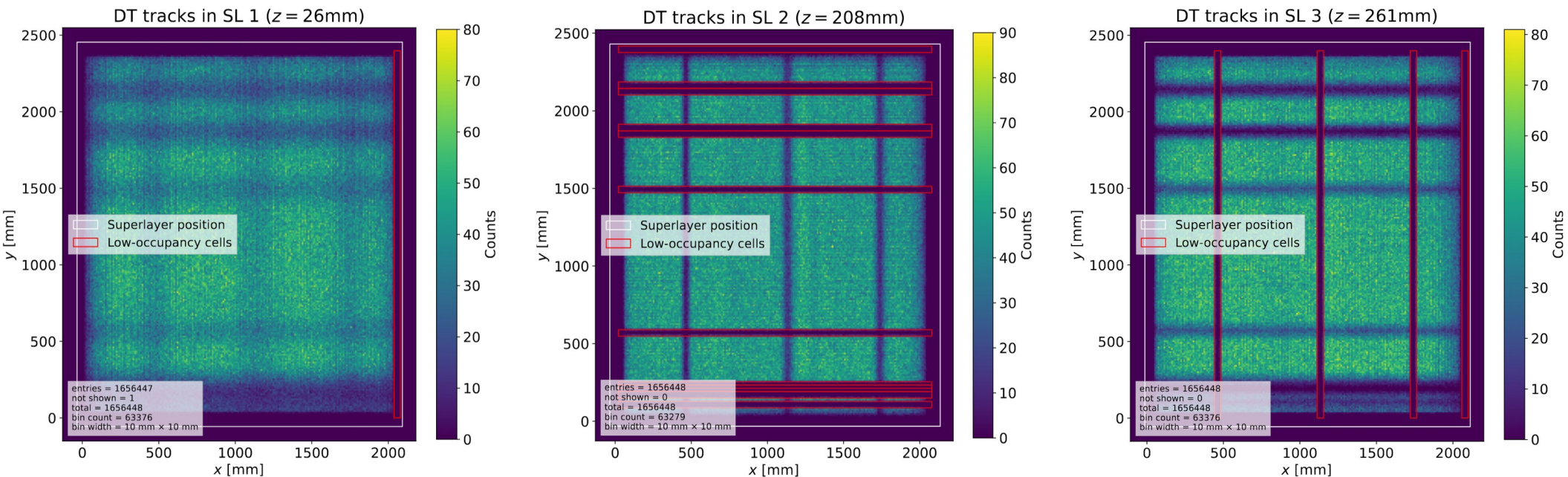
DT global track results

Example reconstructed track:



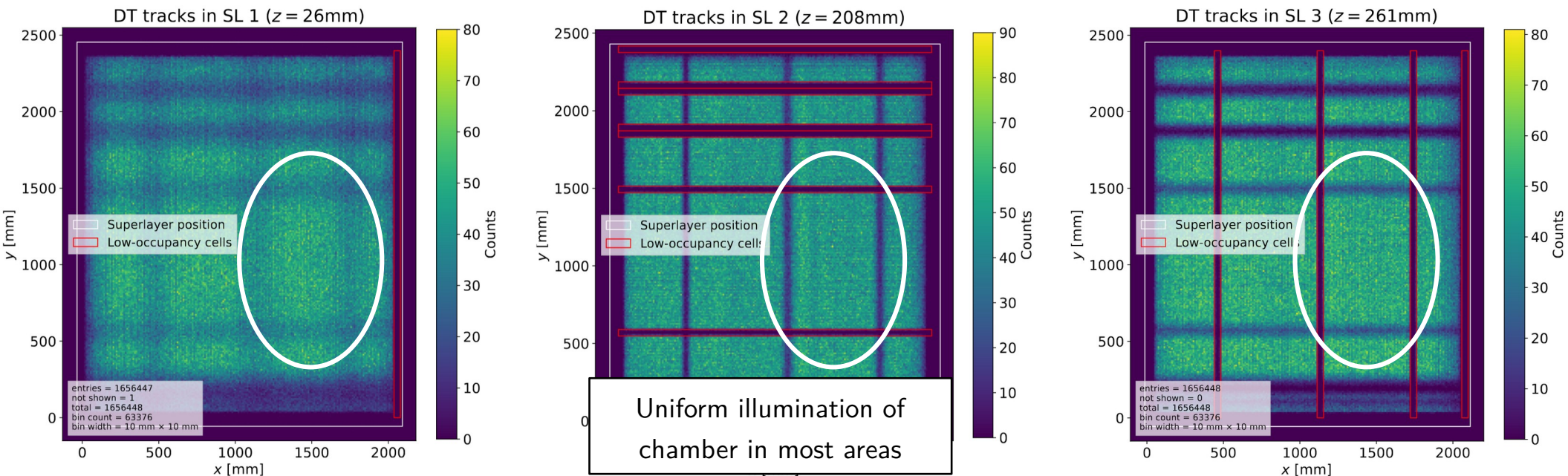
DT global track results

2D histograms of global track hits in superlayers:



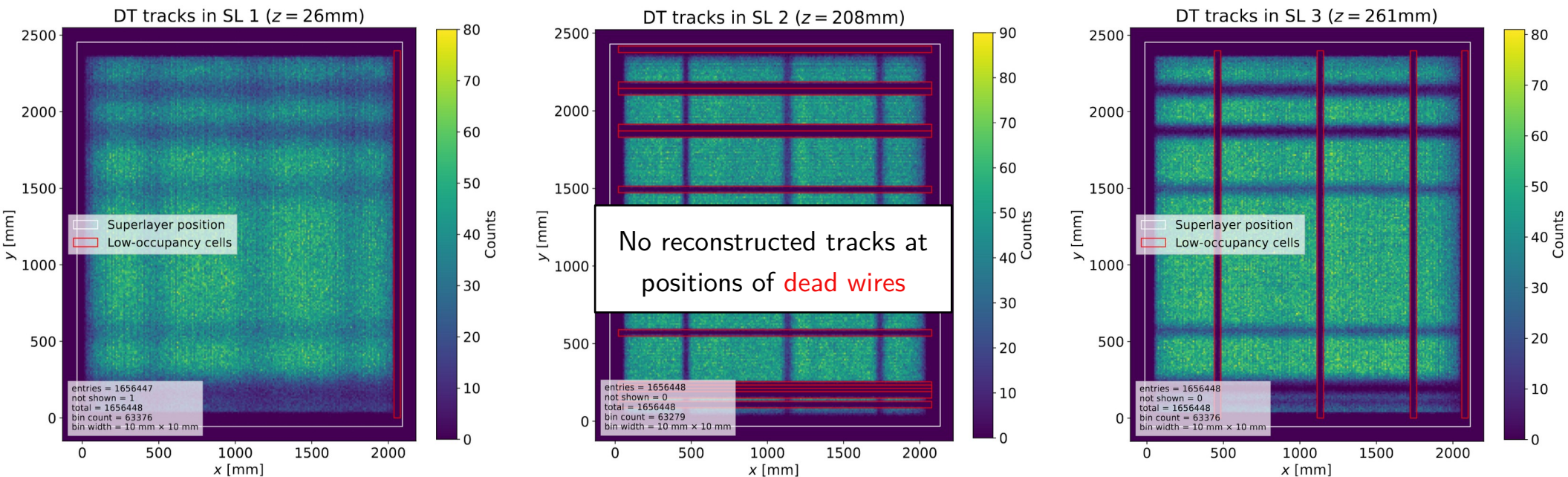
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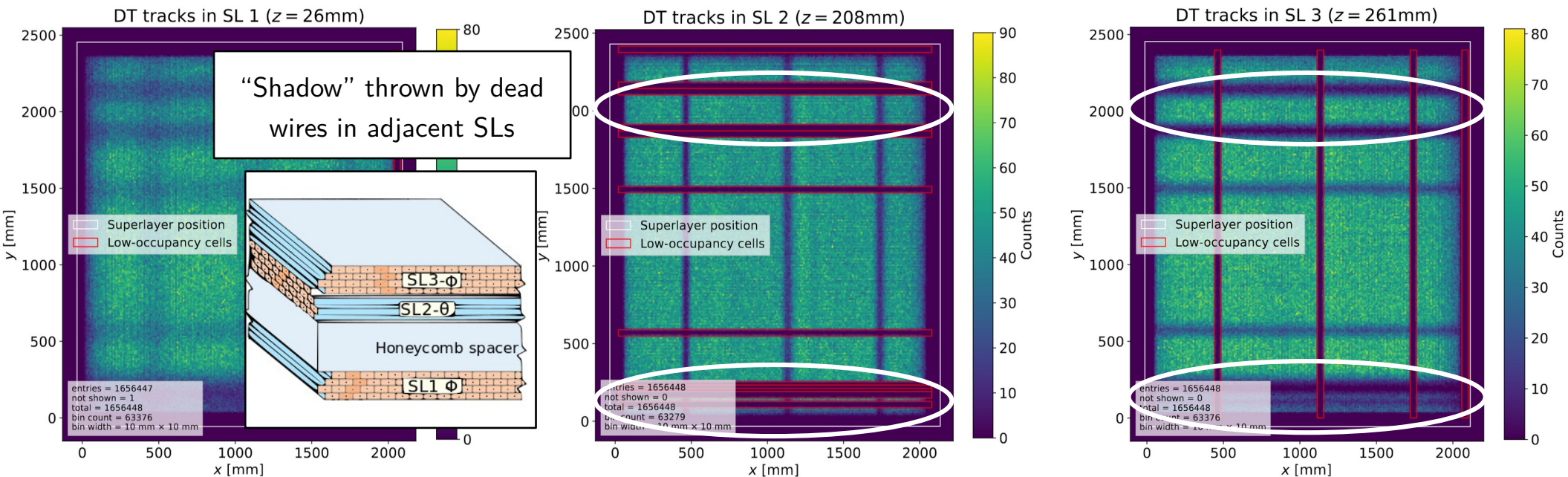
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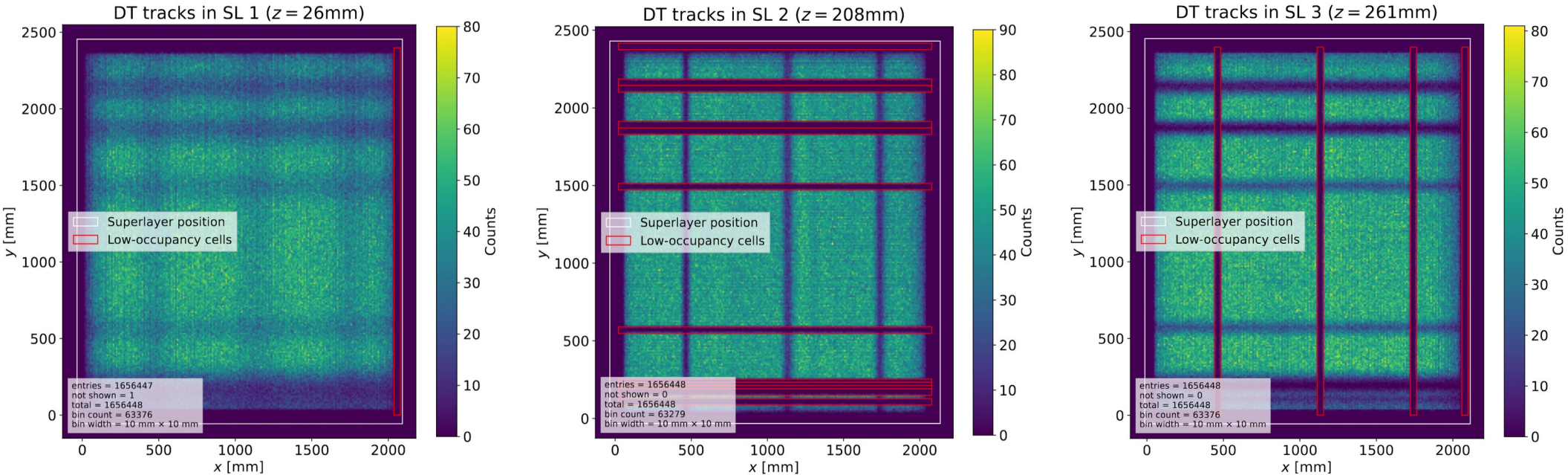
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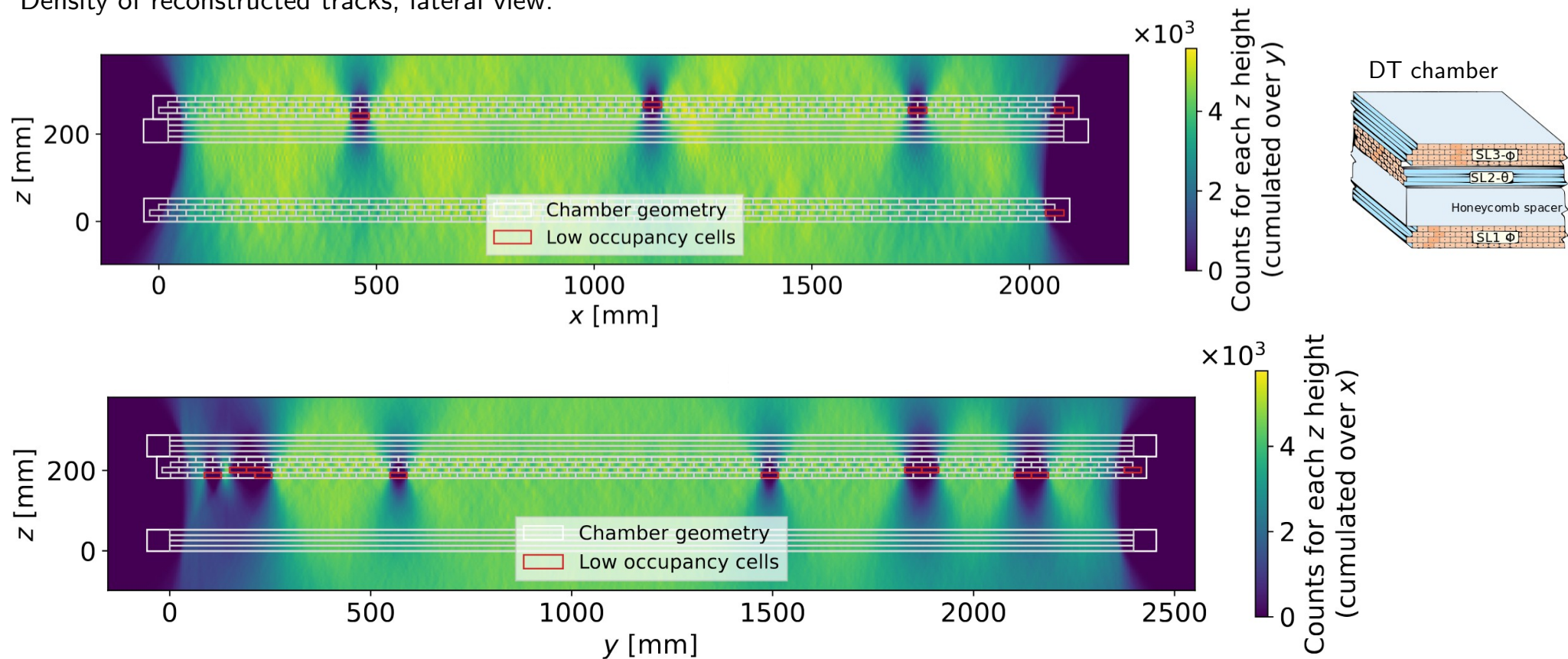
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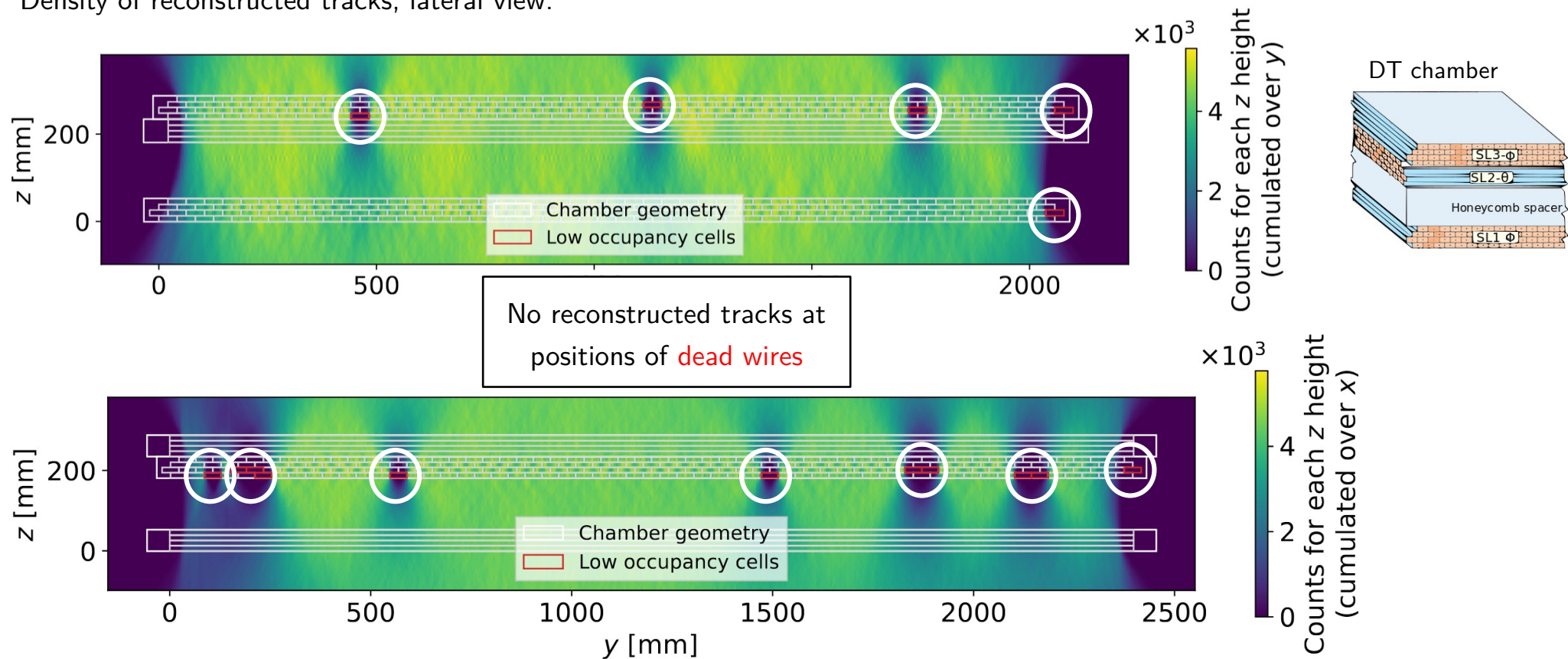
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Density of reconstructed tracks, lateral view:



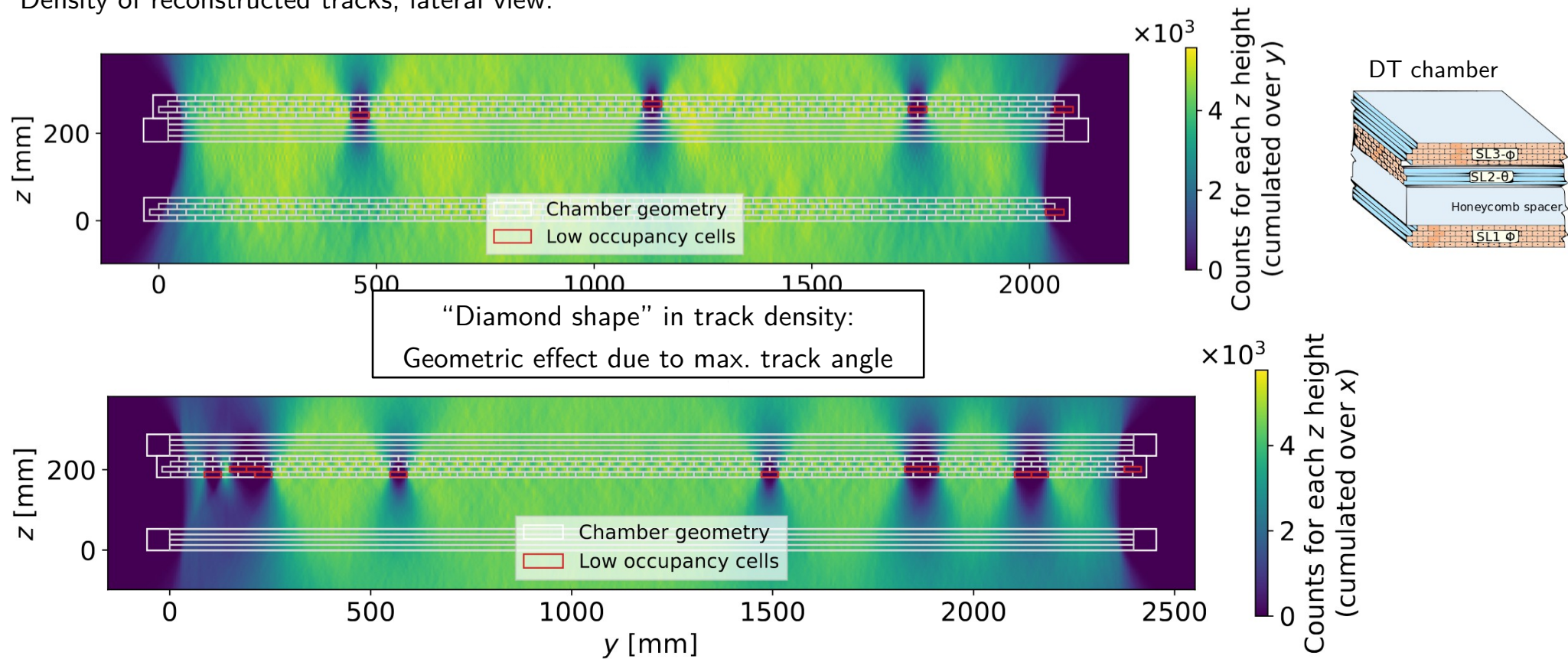
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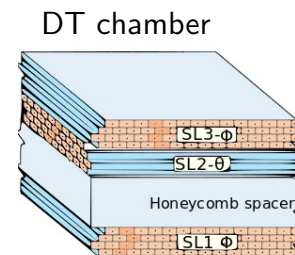
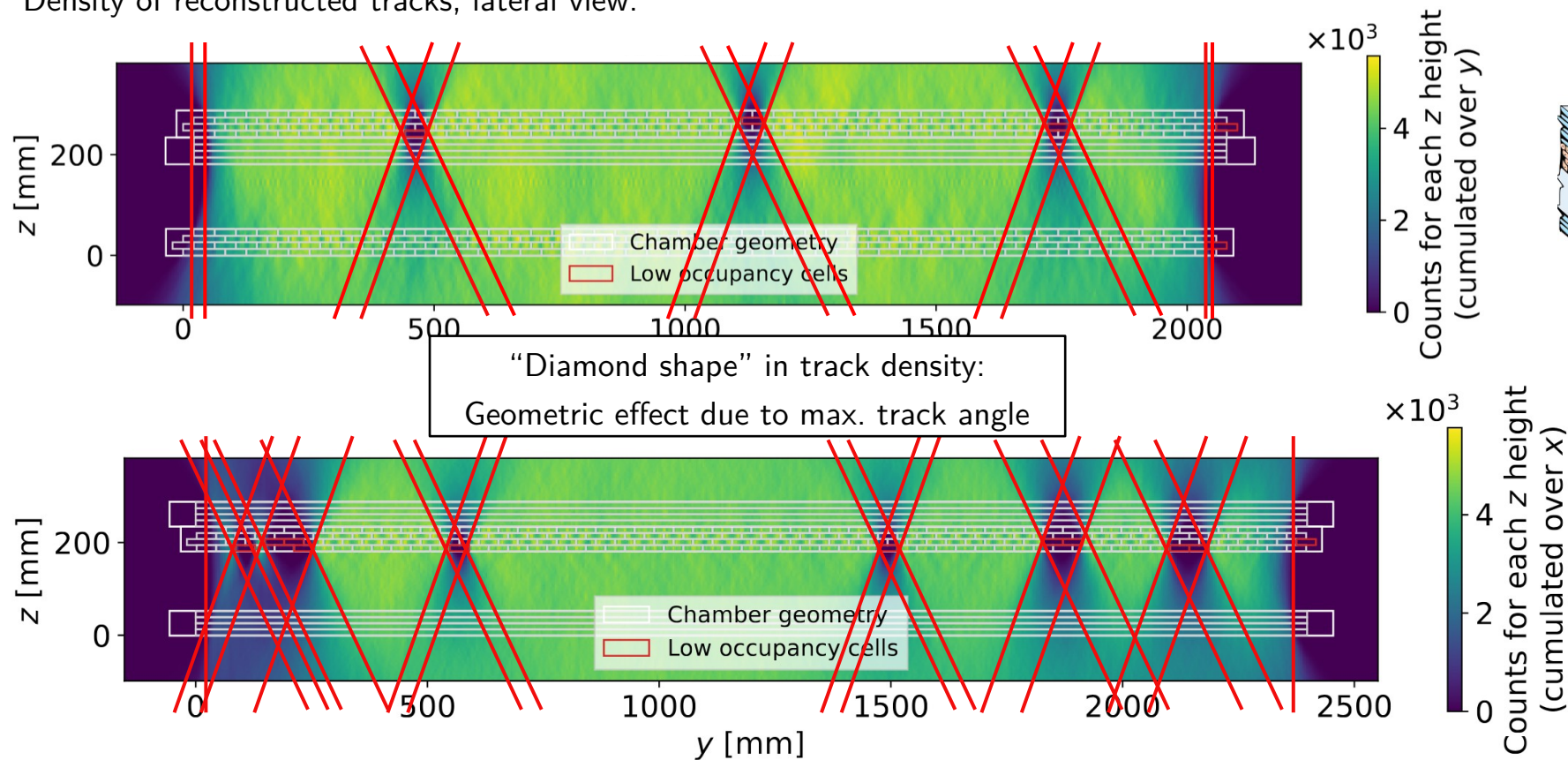
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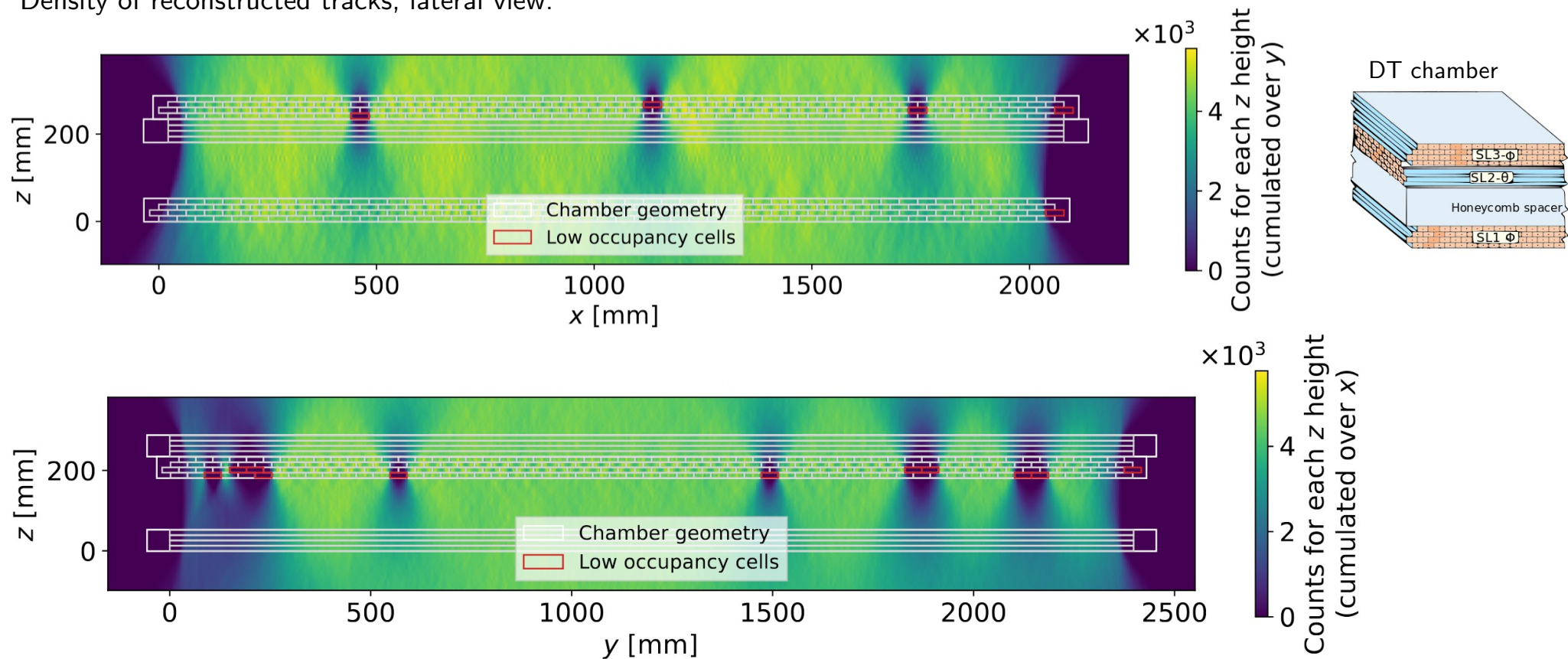
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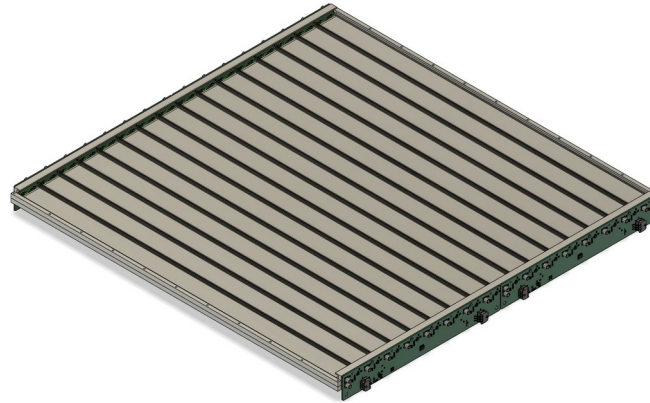
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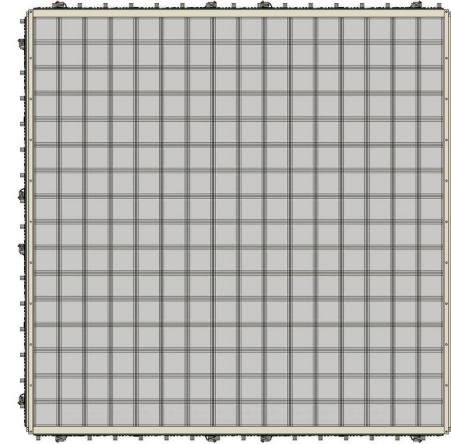
Scintillator strip detector

Scintillator strip detector with SiPM readout

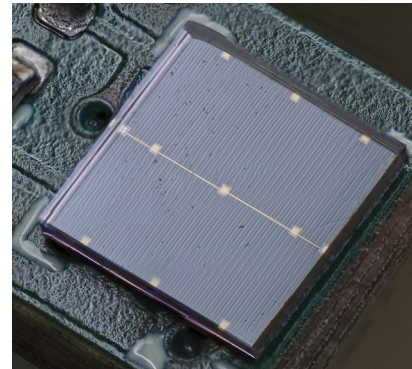
- Scintillator strip detector:
 - Developed in previous thesis projects
 - Plastic scintillator material (EJ-200)
 - Detector size: $50 \times 50 \text{ cm}^2$
 - 2 layers with 16 strips each
 - Strip width: 3 cm
 - Photon readout: 2 SiPMs / strip
- Silicon photomultiplier (SiPM):
 - Solid-state detector with single-photon sensitivity



Scintillator strips



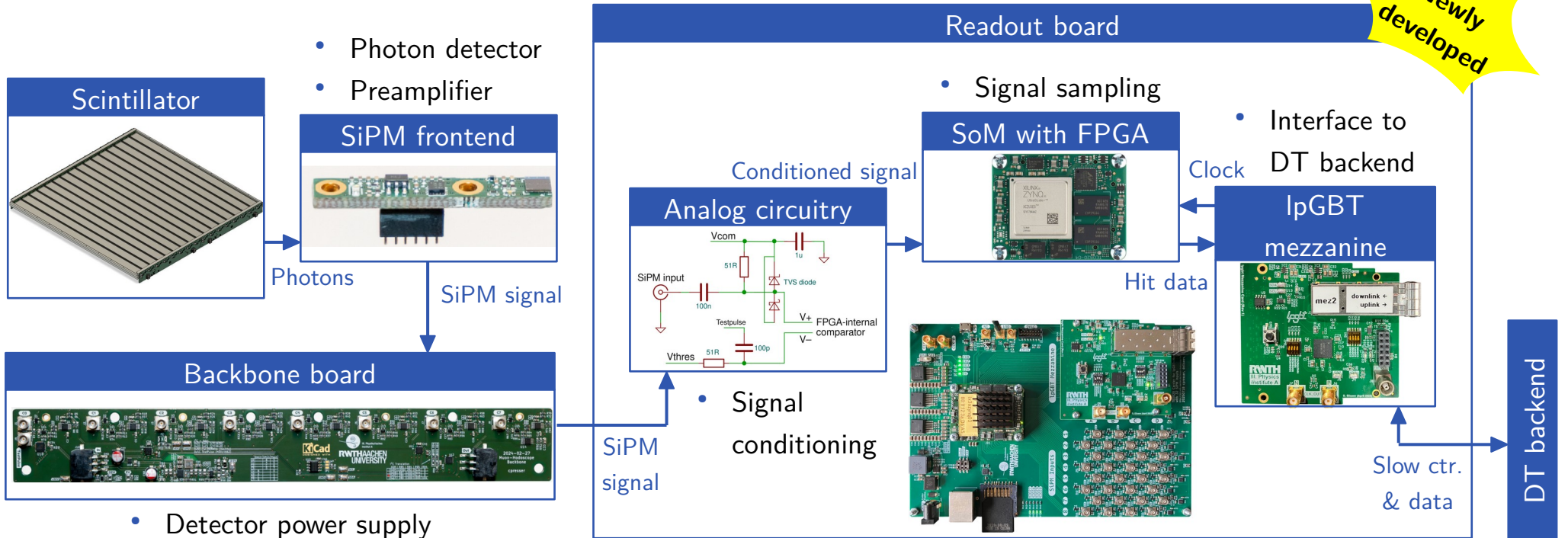
Scintillator top view



Mounted SiPM

Scintillator readout electronics – Overview

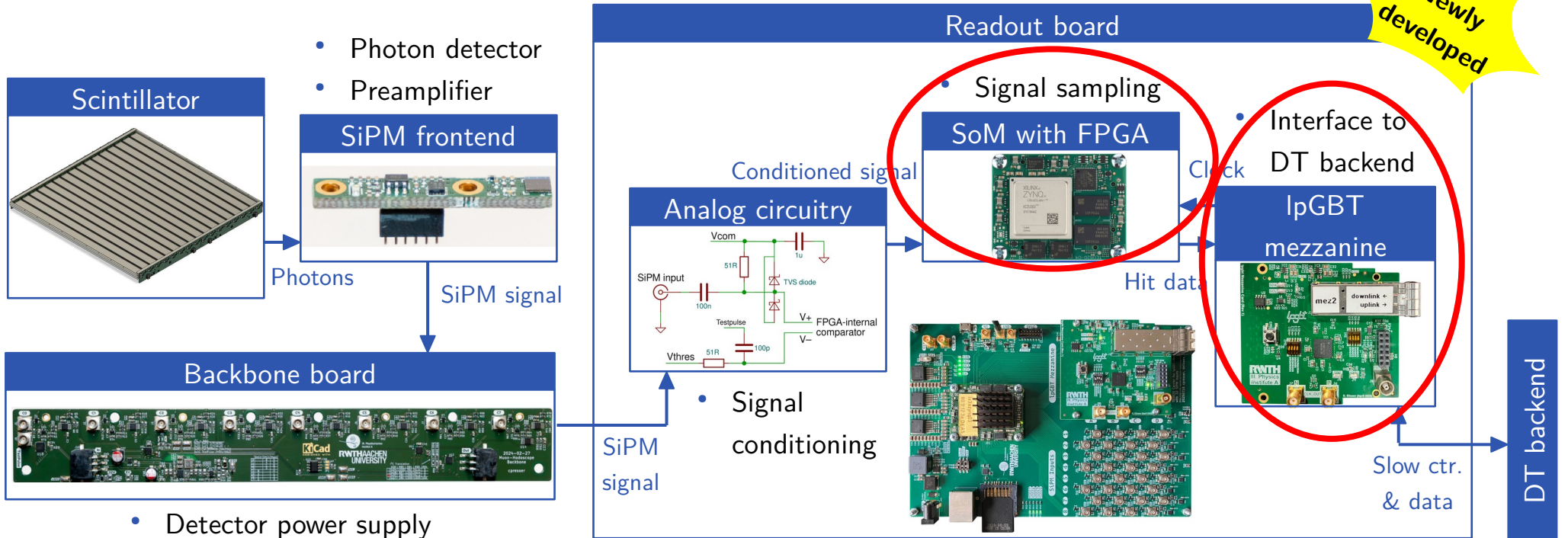
- ▶ Should have capability to interface DT backend



- Data acquisition
- Global clock

Scintillator readout electronics – Overview

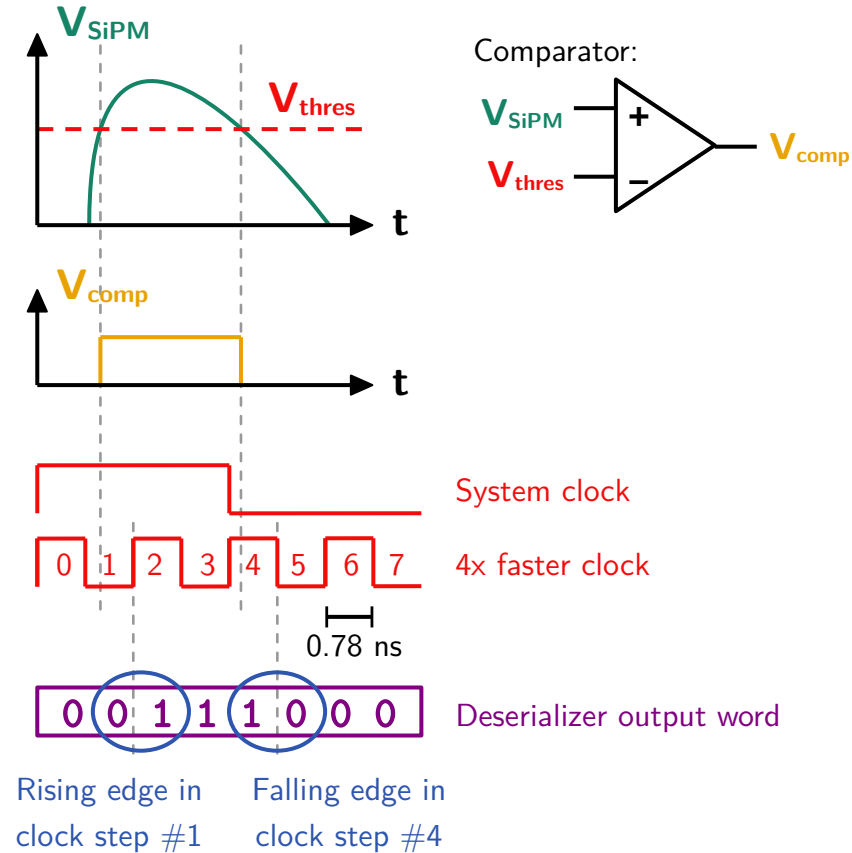
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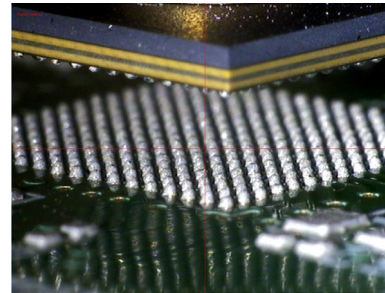
Scintillator readout electronics – Signal sampling

- Receive preamplified analog signal from SiPM frontend
- Discriminator:
 - $V_{thres} = 25 \text{ mV}$ (5 p.e. of SiPM)
 - Implemented with FPGA-internal comparators
- Time-to-Digital Converter (TDC):
 - Implementation in FPGA firmware (using deserializers)
 - Sampling of comparator output with fast clock
 - Detection of edges in resulting bit vector

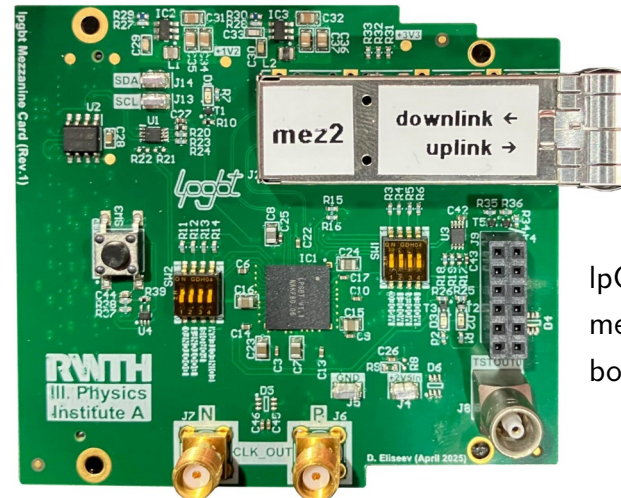


Scintillator readout electronics – lpGBT mezzanine

- Low-Power Gigabit Transceiver (lpGBT):
 - Chip for high-speed bidirectional data transfer
 - Extensively used in Phase-2 Upgrade of CMS
 - Capability to recover synchronous clock from backend
- lpGBT mezzanine:
 - Self-developed board which hosts lpGBT chip
 - FPGA interface: Serial data lines (eLinks)
 - DT backend interface: Optical fiber



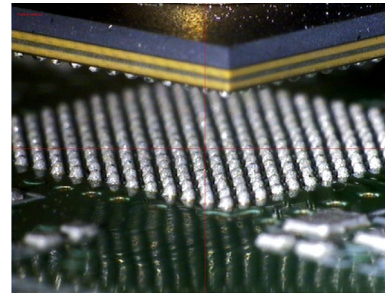
lpGBT chip being soldered onto a board



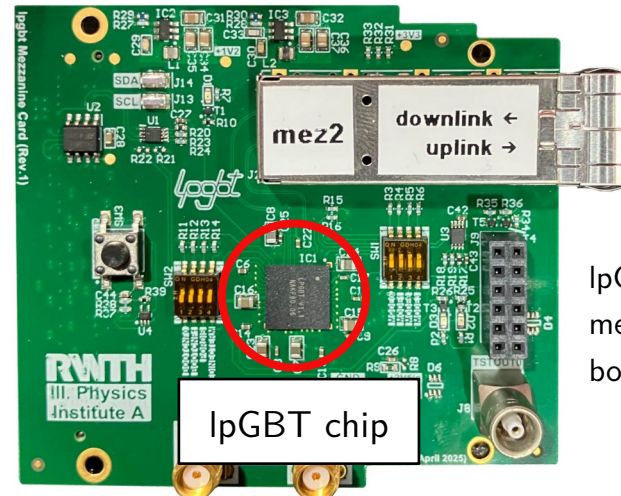
lpGBT mezzanine board

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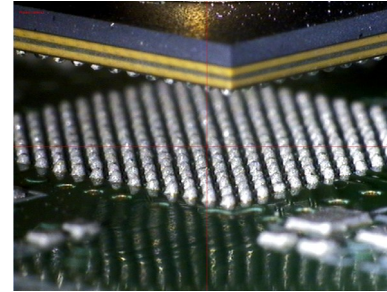
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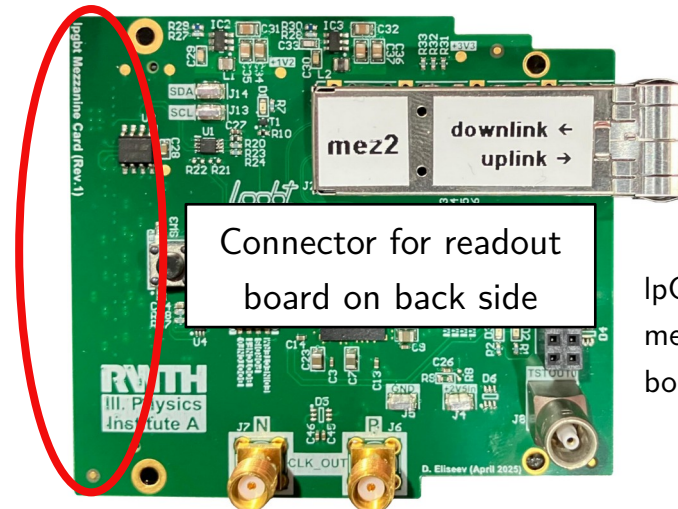
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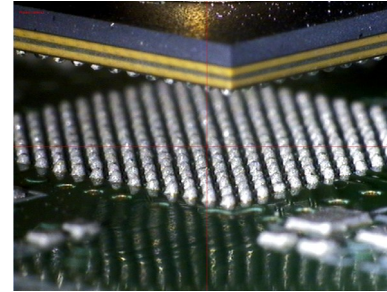
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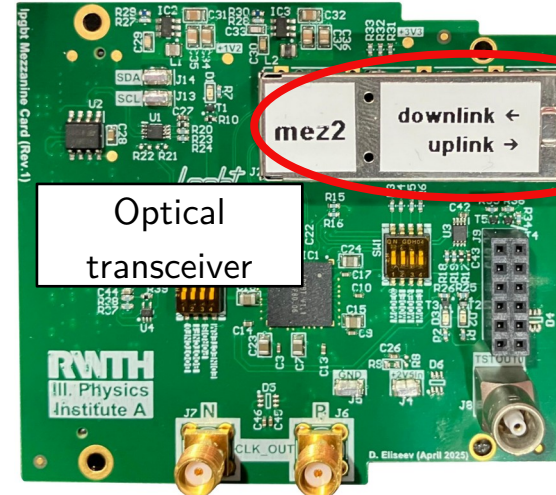
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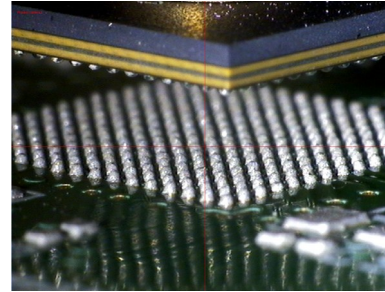
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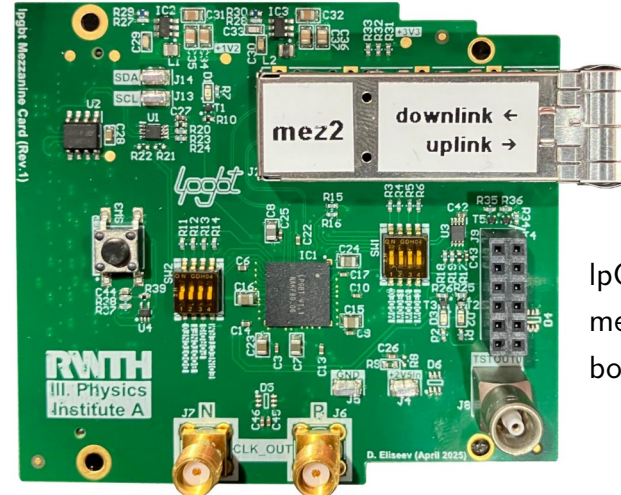
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lpGBT chip being soldered onto a board



lpGBT mezzanine board

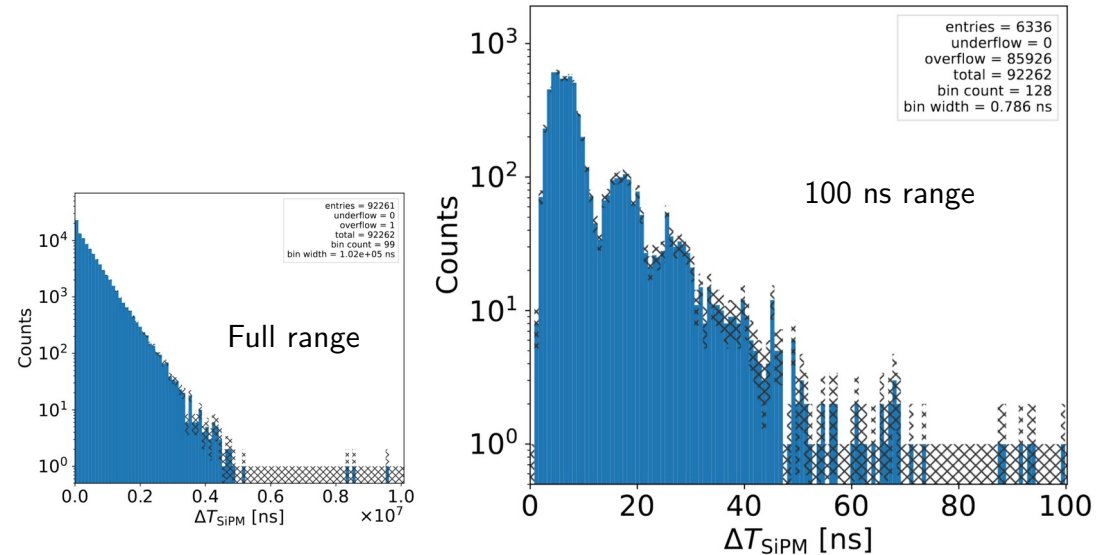
Recorded scintillator strip detector data

SiPM hit data

Without dead time:

- Observe systematic effects in SiPM hit data
 - High **SiPM rate** $O(1 \text{ kHz})$ for selected threshold
 - **Secondary hits** with discrete timing
- ▶ Possible signal reflections or problems with sampling logic
 - ▶ Apply dead time as countermeasure

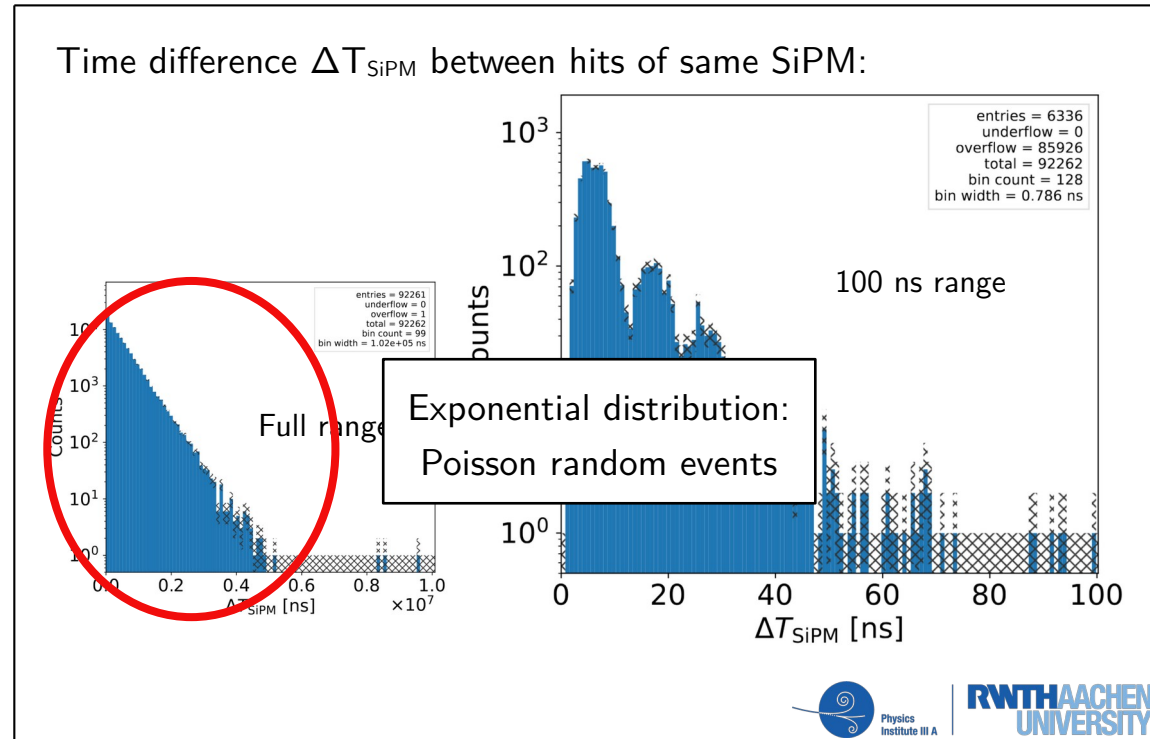
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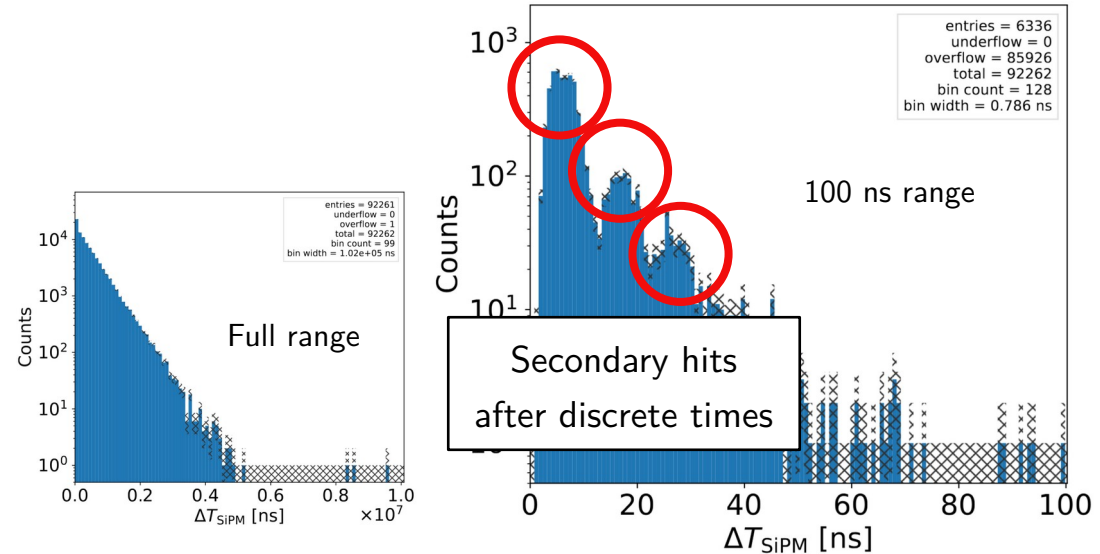


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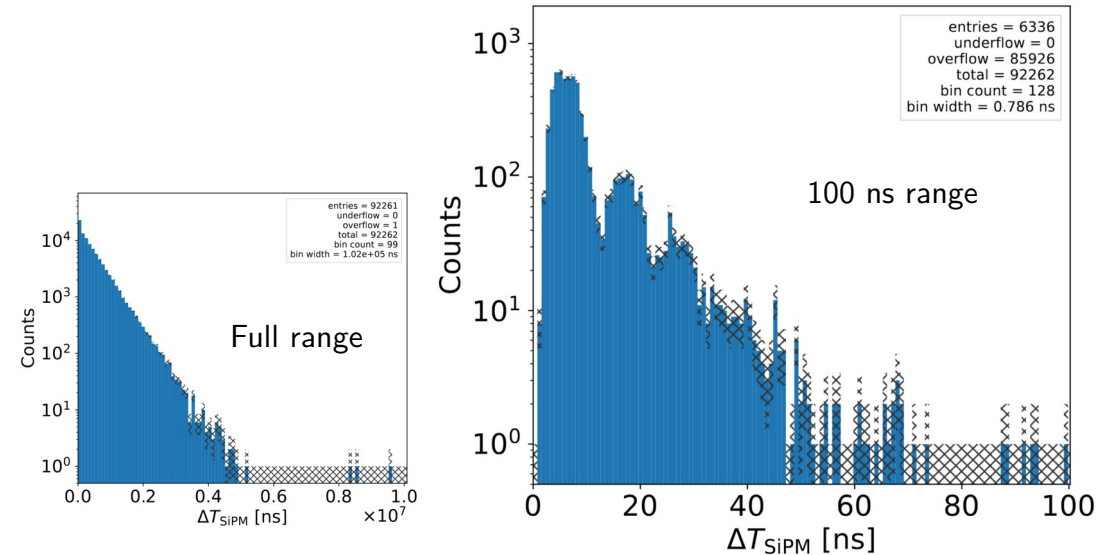


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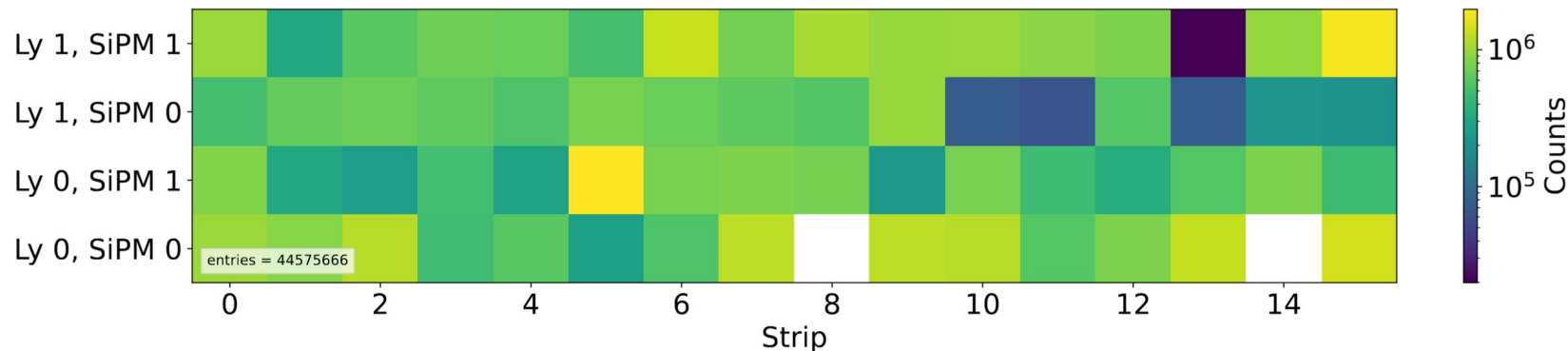
Time difference ΔT_{SiPM} between hits of same SiPM:



SiPM hit data

With 250 ns dead time:

SiPM occupancies:

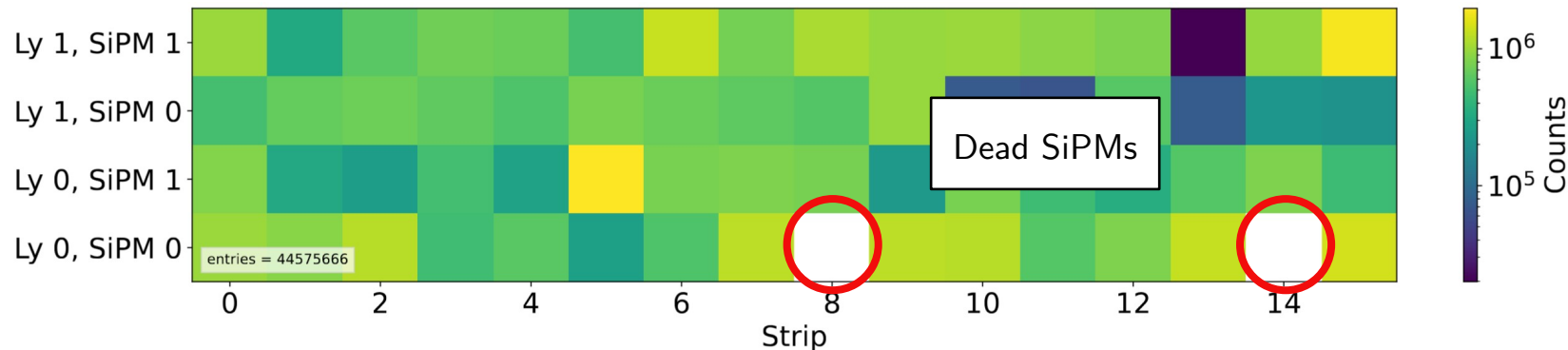


- ▶ Measured SiPM rate: **37 Hz**
 - ▶ Significantly lower rate as without dead time
 - ▶ Rate fluctuations over one order of magnitude
 - Common bias voltage for all SiPMs, i.e. different effective p.e. thresholds
- ▶ **2 dead SiPMs** in detector

SiPM hit data

With 250 ns dead time:

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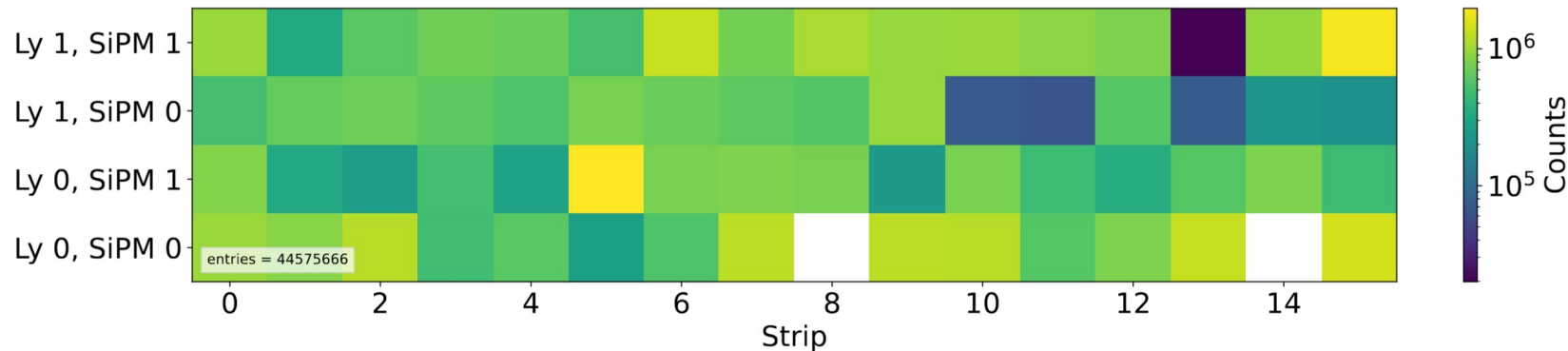


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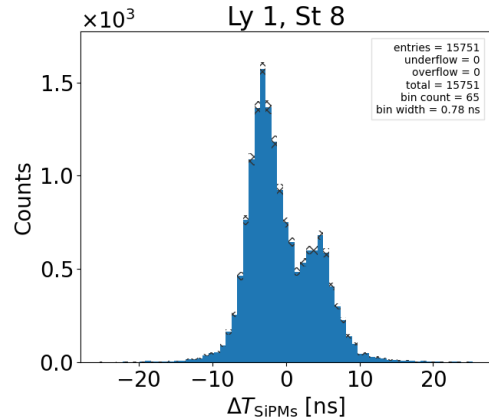


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Scintillator strip coincidences

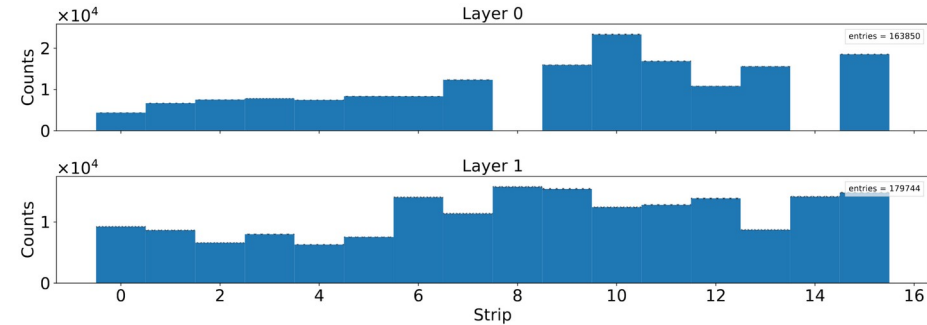
- Evaluate SiPM coincidences for strips
 - Coincidence window: $\Delta T_{\text{SiPMs}} \leq 25 \text{ ns}$
 - Require isolation from other hits in layer

Time difference ΔT_{SiPMs} of coincidence:



- ▶ **Contained coincidence timing $O(15 \text{ ns})$**
- ▶ Double-peak structure observed
 - ▶ Needs further investigation, possibly due to sampling logic

Obtained strip occupancies:

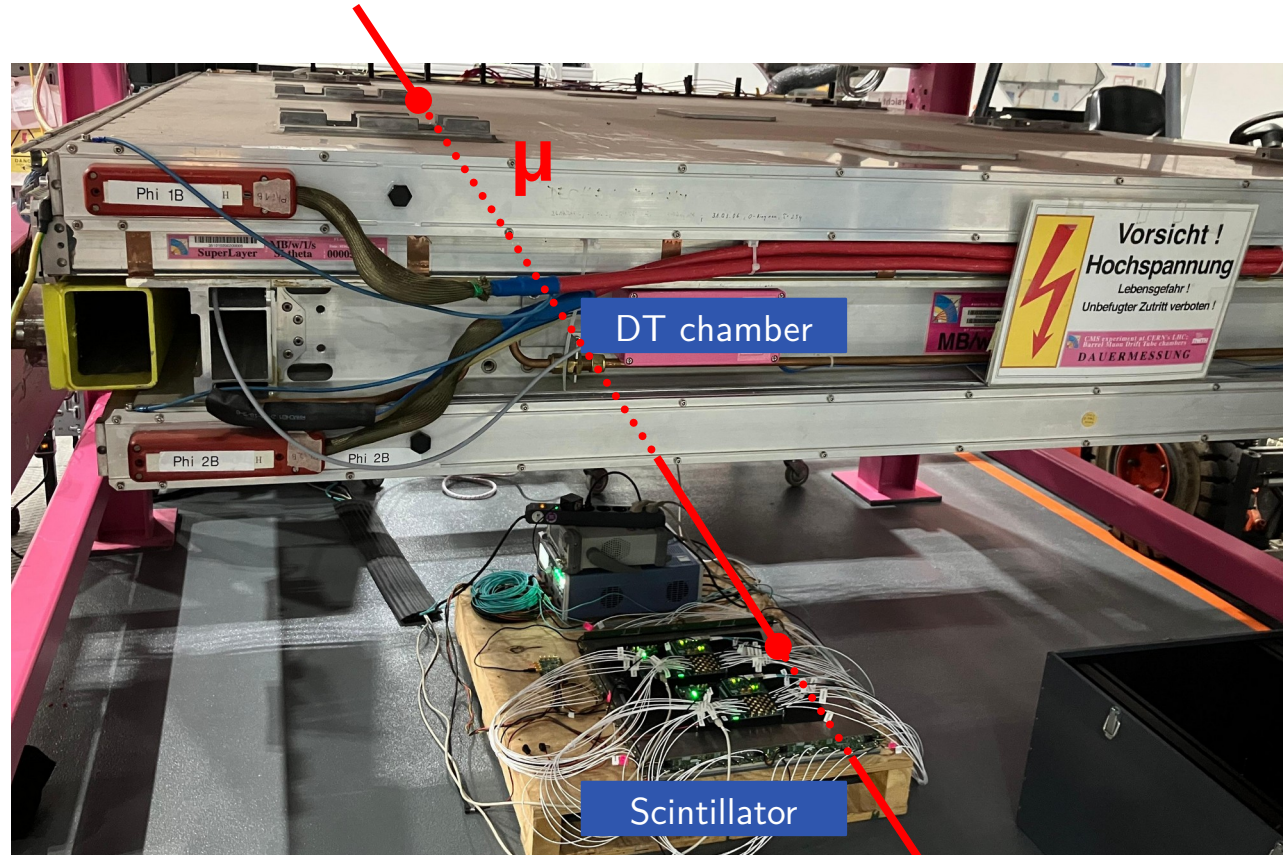


- ▶ Measured strip rate: **$O(0.5 \text{ Hz})$**
 - ▶ Strip rate fluctuations of factor ~ 5
 - ▶ **2 dead strips** from dead SiPM propagation

Combination of detector data

Combined measurement setup

- Place scintillator below DT chamber at cosmic test stand
- Attempt combined data analysis of DT tracks & scintillator strip hits

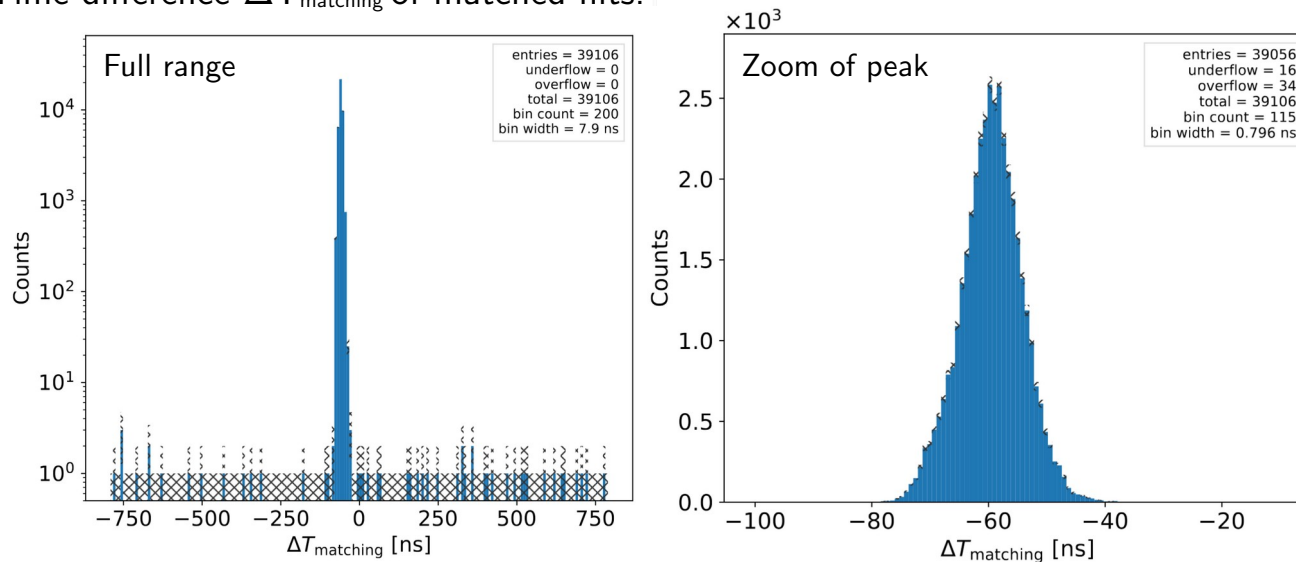


Temporal matching of DT tracks & scintillator hits

- Matching of detector data in time:
 - Require DT track & scintillator strip hit coincidence:
 $\Delta T_{\text{matching}} = T_{\text{scint}} - T_{\text{DT}} \leq 780 \text{ ns}$ (very conservative)
 - Remove ambiguities while matching

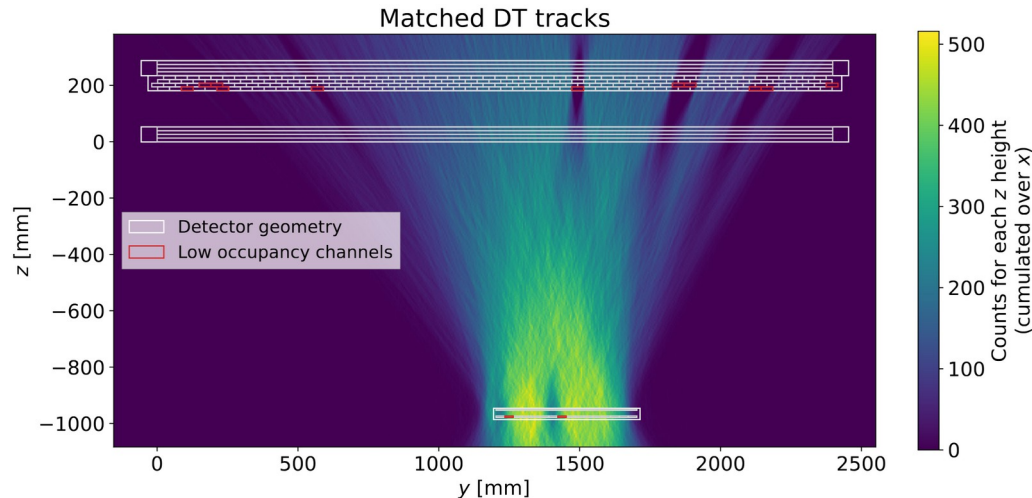
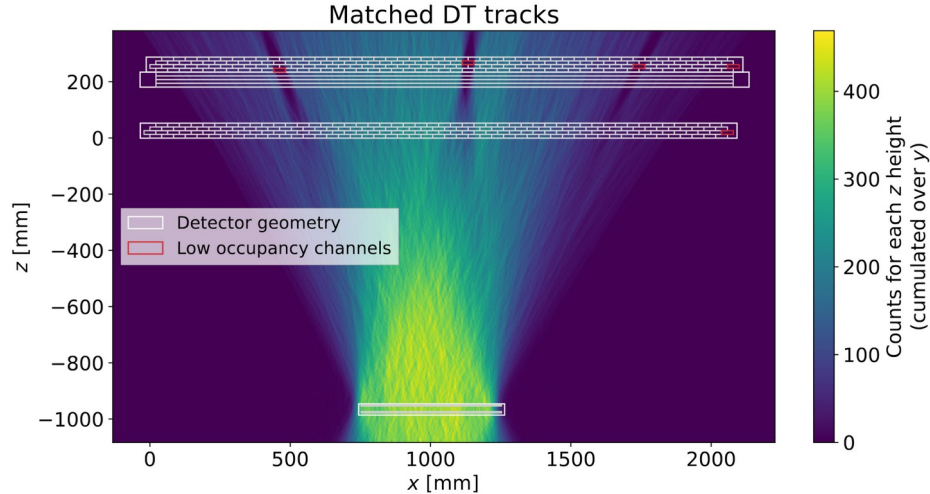
- ▶ Matched hit rate: **2 Hz**
 - ▶ DT track rate scaled to scintillator: 4 Hz
 - ▶ Scintillator layer rate: 8 Hz

Time difference $\Delta T_{\text{matching}}$ of matched hits:



- ▶ **Achieve synchronization with $O(12 \text{ ns})$ precision !**
 - ▶ Includes all timing uncertainties of readout & reconstruction

Independent cross-check of DT track reconstruction

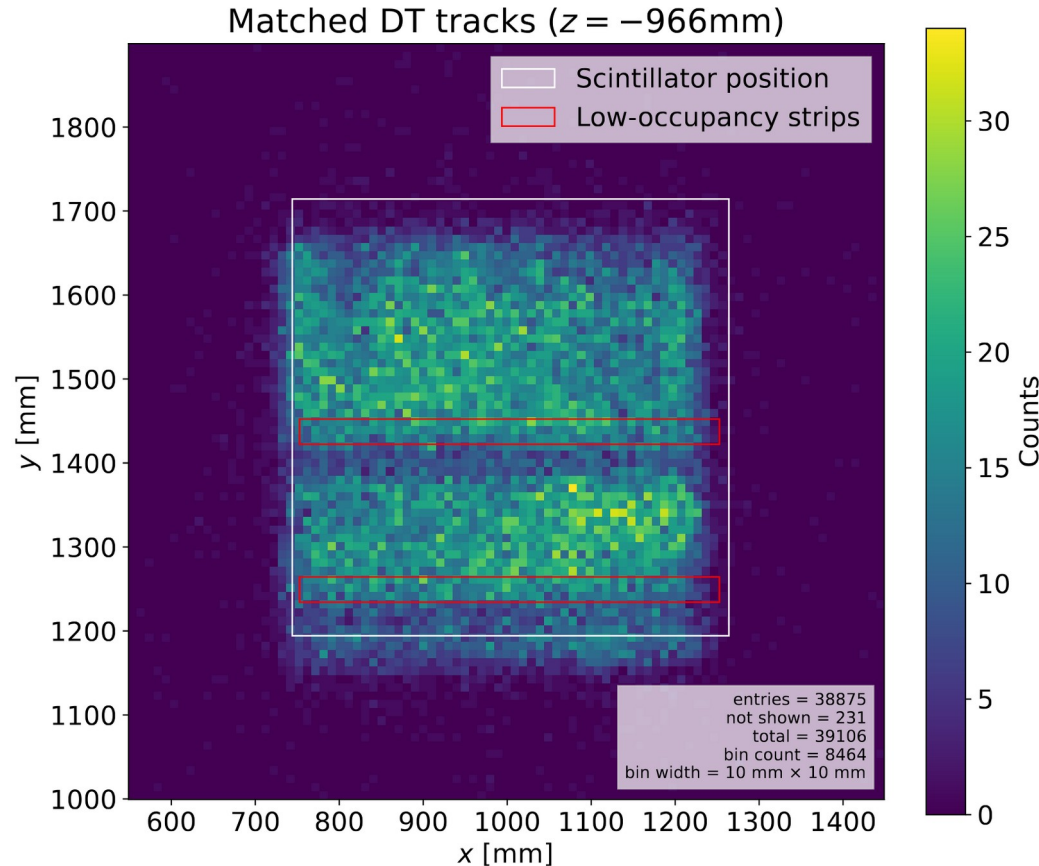


- Cross-check of reconstructed DT tracks
 - Analyze track position of tracks w.r.t. scintillator
 - Only matched in time, therefore unbiased
- ▶ **Matched tracks are intersecting scintillator as expected !**
 - ▶ DT track reconstruction seems to work properly
 - ▶ Visible effect of DT cells & scintillator strips
- ▶ Need for proper detector alignment

Density of matched reconstructed tracks, in lateral view.

Independent cross-check of DT track reconstruction

2D histogram of matched tracks in scintillator:

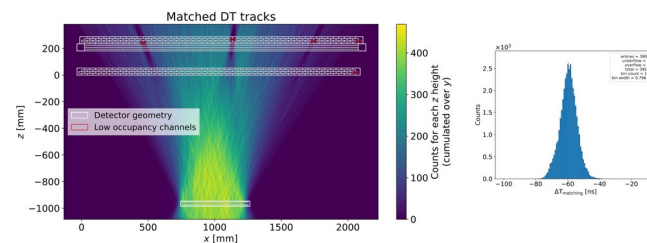
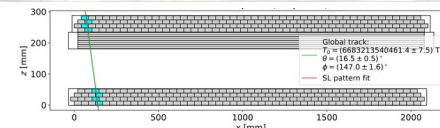


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 - ▶ Visible effect of DT cells & scintillator strips
- ▶ Need for proper detector alignment

Summary & outlook

Summary & outlook

- DT chamber at cosmic test stand:
 - Successful installation & commissioning of CMS DT Phase-2 electronics
 - Implementation of triggerless track reconstruction algorithm
- Scintillation strip detector:
 - Developed new readout system & firmware (featuring lpGBT chip)
 - Observed systematic effects in SiPM readout and established mitigation with dead time
- Combined analysis:
 - Detector synchronization $O(12\text{ ns})$ was achieved, including all timing uncertainties
 - Positive independent cross-check of DT track reconstruction



Outlook:

- ▶ Further investigate systematics in SiPM readout system
- ▶ Optimize triggerless DT reconstruction algorithm / Investigate possibility to use online DT trigger algorithm
- ▶ Provide detector alignment method
- ▶ Gained experience can be used to convert DT chamber to cosmic test stand for other detectors (beyond the scintillator)

Thank you for your attention!

Questions and comments are of course welcome.

Thank you!

I would like the following (incomplete) list of people who made this thesis possible:

- Prof. Dr. Alexander Schmidt for the opportunity to write my thesis at his institute
- Apl. Prof. Dr. Oliver Pooth for kindly agreeing on being the second corrector
- Dr. Markus Merschmeyer for his support and feedback
- Dr. Dmitry Eliseev for supervising this thesis and his tireless support
- Carsten Presser for contributing several important aspects
- Prof. Dr. Thomas Hebbeker for his interest and the helpful suggestions

Backup

- 1) Theory
- 2) DT chamber description & data analysis
- 3) Scintillator description & data analysis
- 4) Combined data analysis
- 5) Simplified Monte Carlo simulation of cosmics

Backup (Pt. 1)

- Theoretical background
- Principles of particle detection

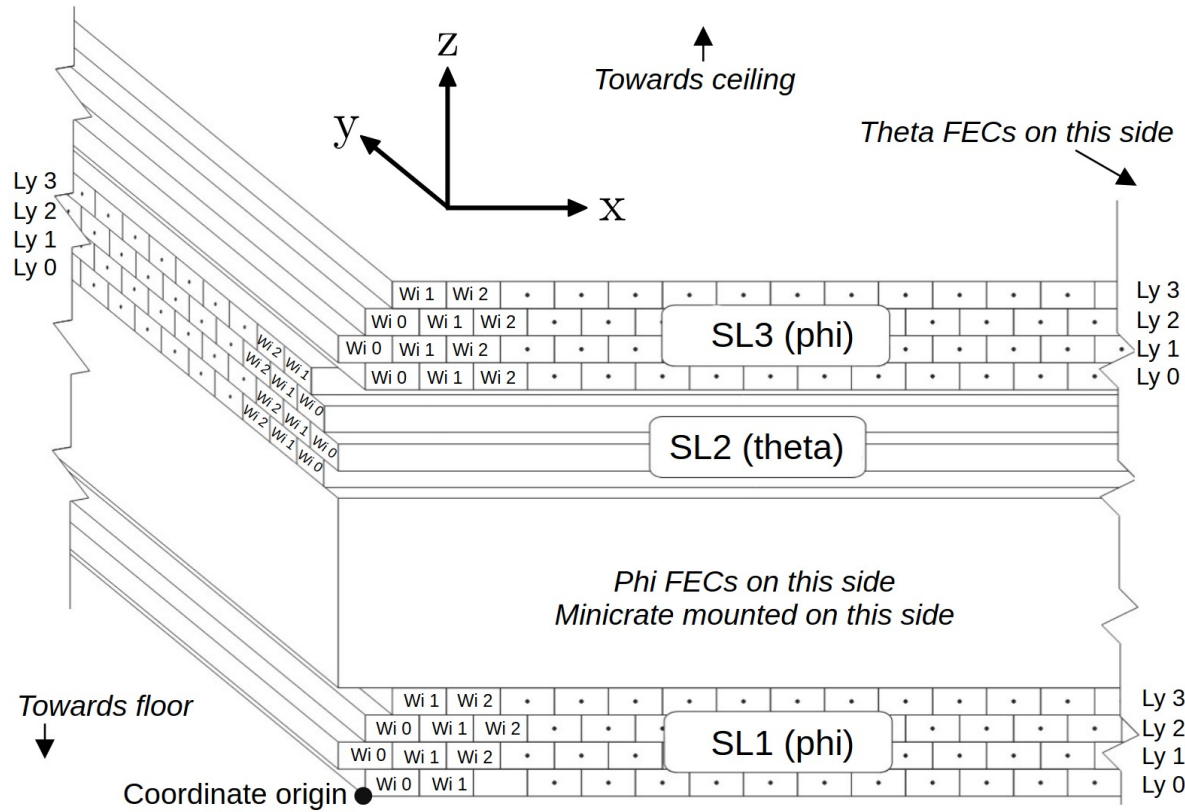
Atmospheric muons

- Cosmic rays interact with Earth's atmosphere, muons originate from decay of light hadrons
- Vertical muon flux: $\Phi = 70 \text{ m}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$
- Zenith angle dependence: $\Phi(\theta) \sim \cos^2 \theta$

Backup (Pt. 2)

- CMS DT system
- Triggerless DT data analysis

DT global coordinate frame



DT readout electronics

DT chamber frontend board:

- Signal discrimination: 35 mV threshold
- Output of digital hit pulse

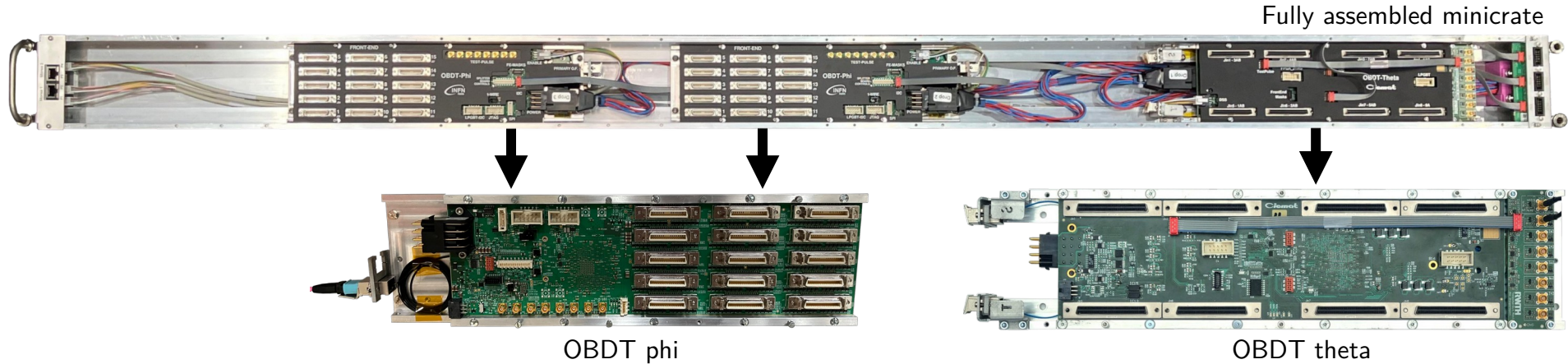


Frontend board

DT readout electronics

Minicrate (on-chamber readout electronics):

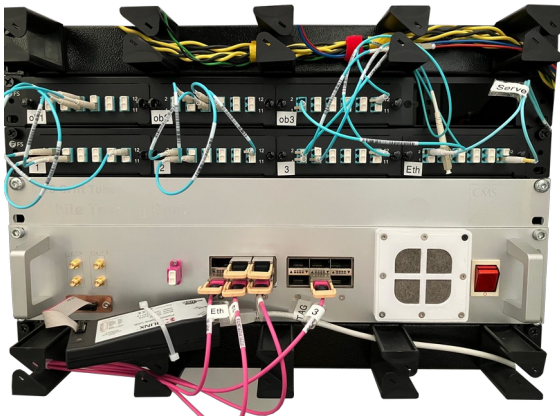
- On-board Electronics for Drift Tubes (OBDDT) boards in two flavors (phi & theta)
 - Hit digitization with 0.78 ns granularity
 - Data transmission over 10.24 Gb/s optical links



DT readout electronics

Backend system:

- Prototype backend hardware: “Slow control & timing box” (SC&T box)
 - No online trigger algorithms, only stores raw DT hits
- Actual CMS DT Phase-2 backend system not available yet

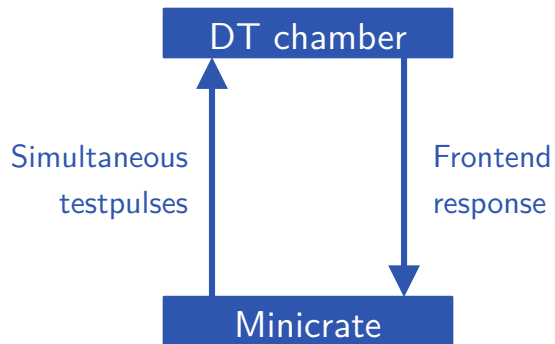
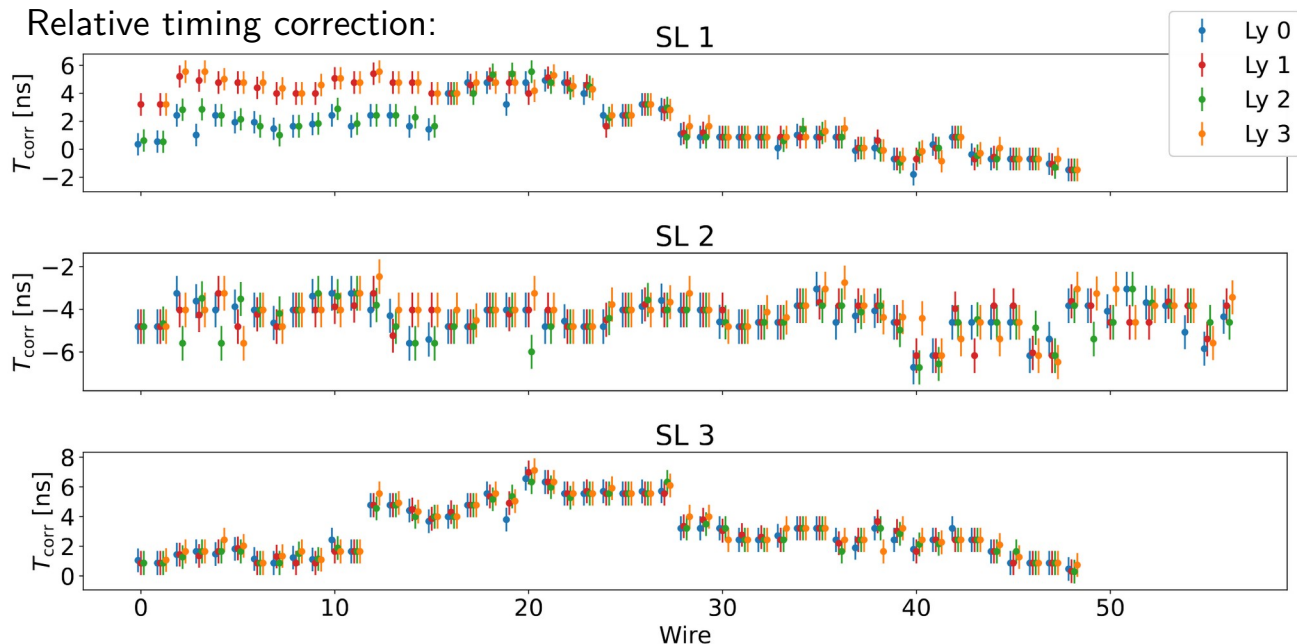


Optical fiber patch panel

SC&T box backend

DT timing calibration with testpulses

- DT reconstruction relies on cell hit timestamps with $O(\text{ns})$ precision
 - Relative timing calibration is necessary
- Minicrate generates simultaneous testpulses
 - Well-defined timing of sent testpulse
 - Record response timestamp and estimate relative correction T_{corr}

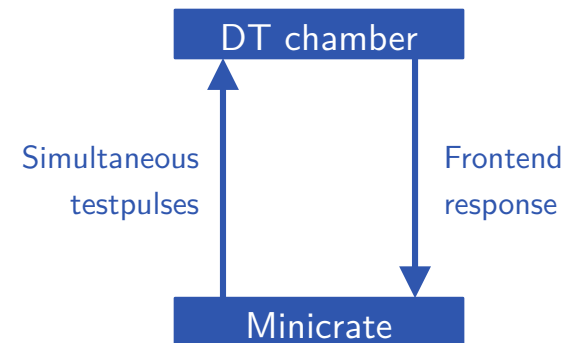
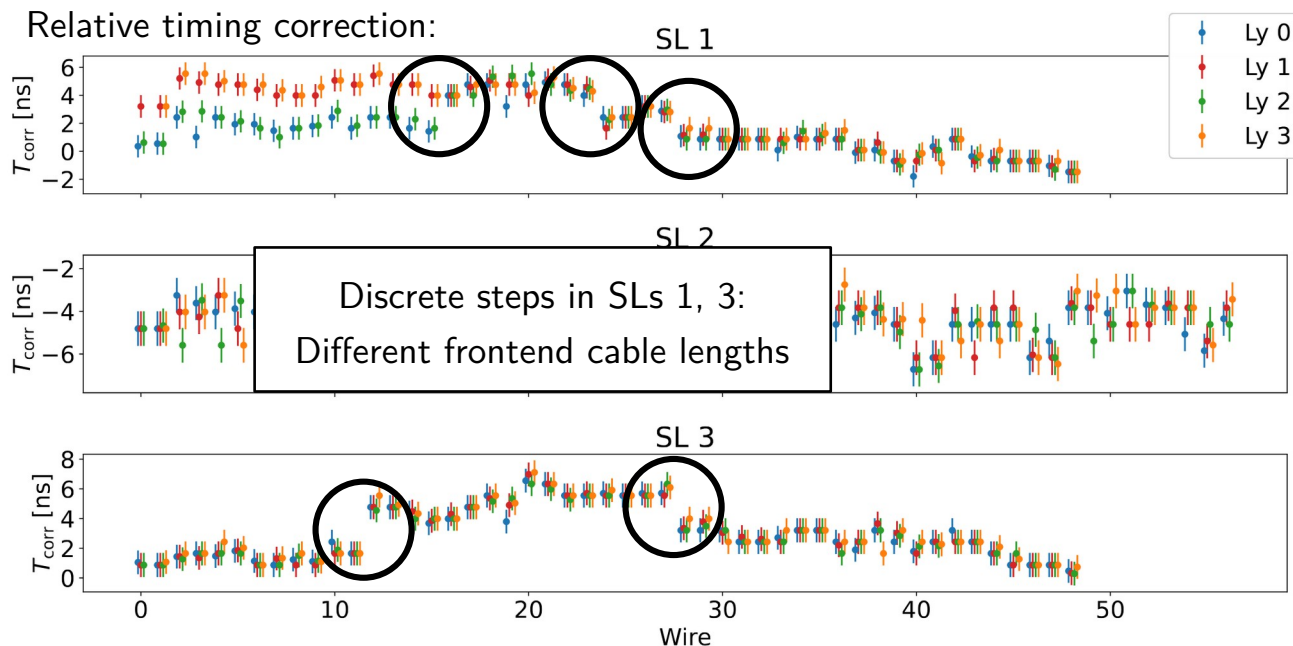


$$T_{\text{corr}} = T_{\text{response}} - \langle T_{\text{response}} \rangle$$

- ▶ Obtain DT timing corrections in the expected range (< 10 ns)
- ▶ Are applied in offline data analysis

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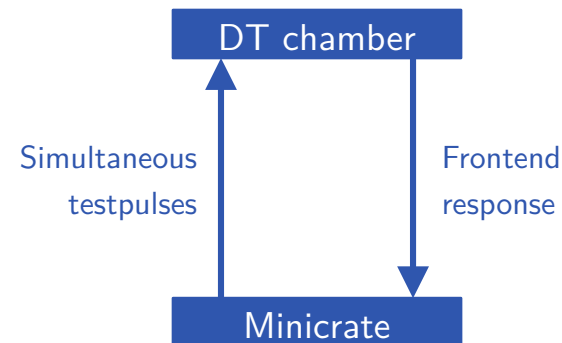
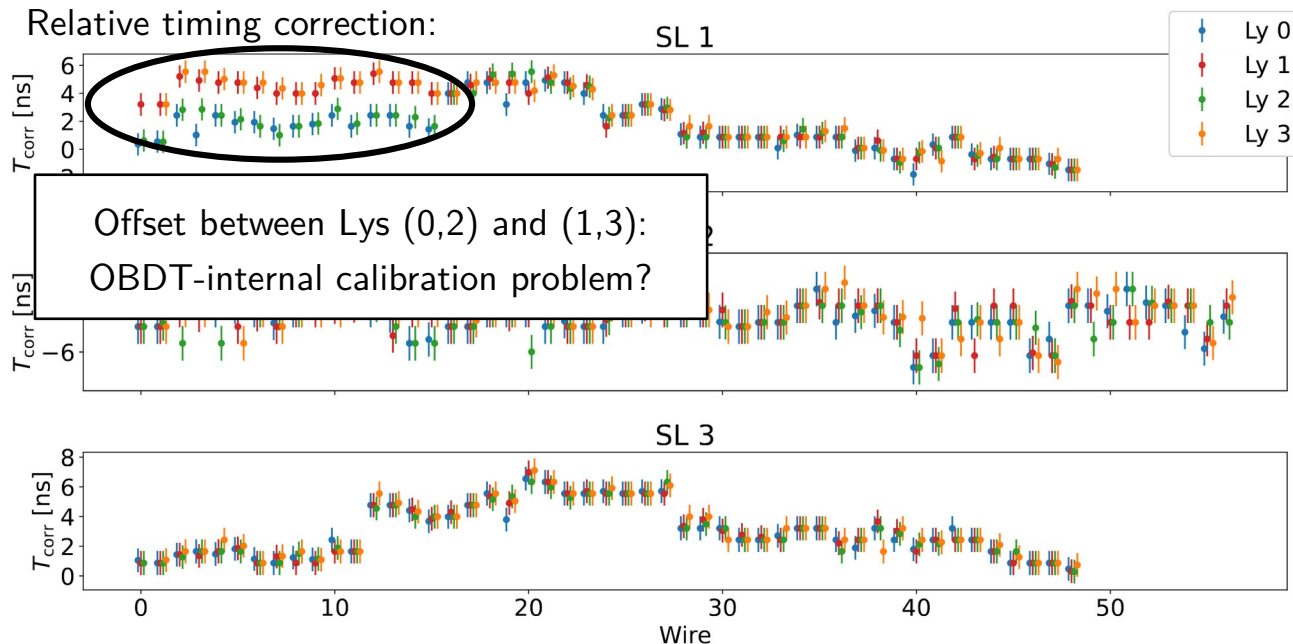


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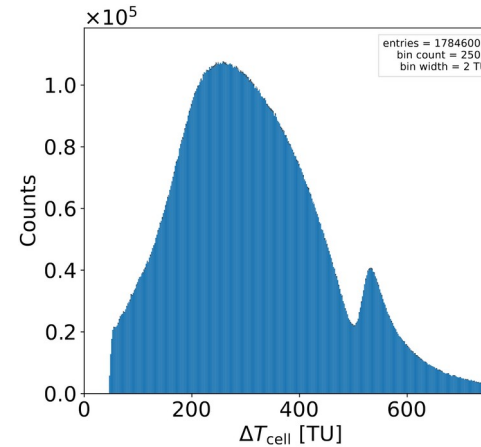
DT secondary hits

- Time difference between consecutive hits of same cell reveals presence of secondary hits
 - Remove Poisson background events
 - Observe double peak structure in < 500 ns range
- Similar structure was observed in CMS publications

Physical cause:

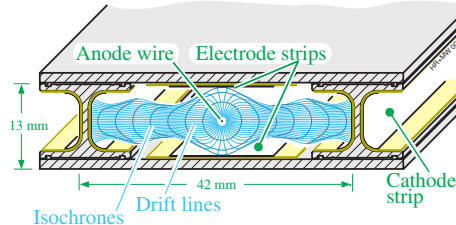
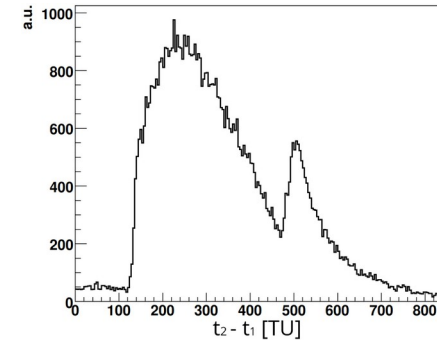
- Photoeffect leads to secondary ionization
 - Photons from avalanche near anode wire create photoelectrons at electrodes
 - Peak 1: Field shaping electrodes
 - Peak 2: Cathode electrodes ($t_{\text{drift}} \approx t_{\text{drift,max}}$)

Time difference between hits:



1 TU (timestamp unit) = 0.78 ns

Comparison with
CMS-NOTE-2008-003:



- ▶ Apply individual cell dead time of 468 ns to suppress secondary hits in the recorded dataset

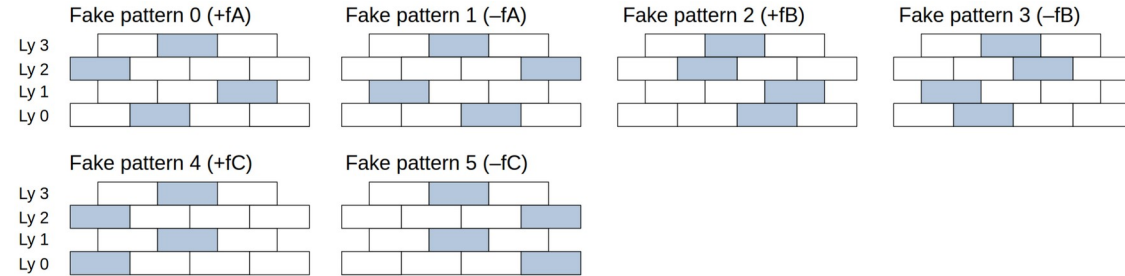
Estimation of drift velocity: $\mathbf{v_{drift} = d_{half-cell} / (t_{peak\ 2} - t_{initial}) \approx (50.5 \pm 0.2) \mu m/ns}$

- ▶ Doubts about precision: Need detailed simulation to understand all involved effects
- ▶ Therefore continue analysis with CMS reference value: $v_{\text{drift}} = 54.5 \mu\text{m/ns}$

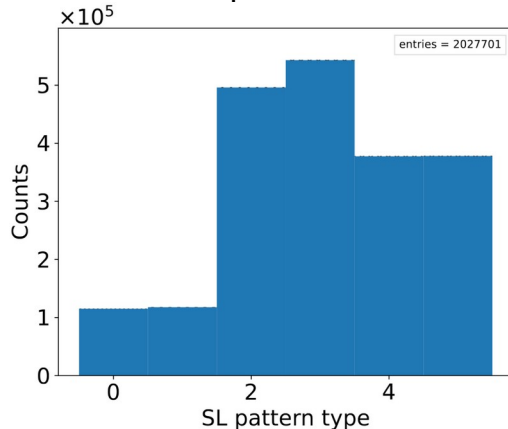
DT superlayer fake patterns

- Analyze influence of noise hits with fake patterns

Superlayer fake pattern geometry:



Pattern occupancies:

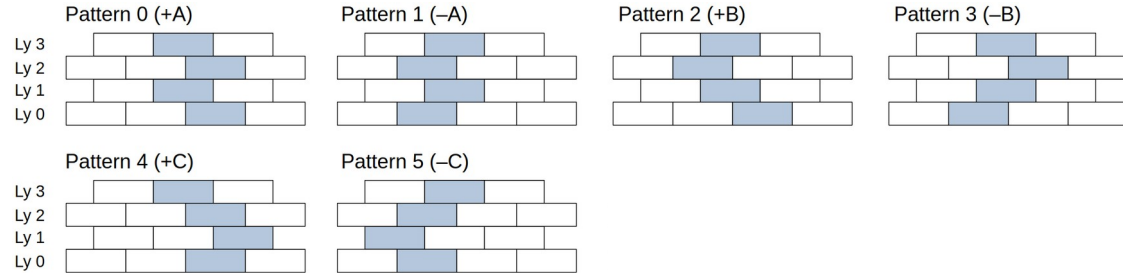


- Observe cumulated pattern rate of 35 Hz / superlayer

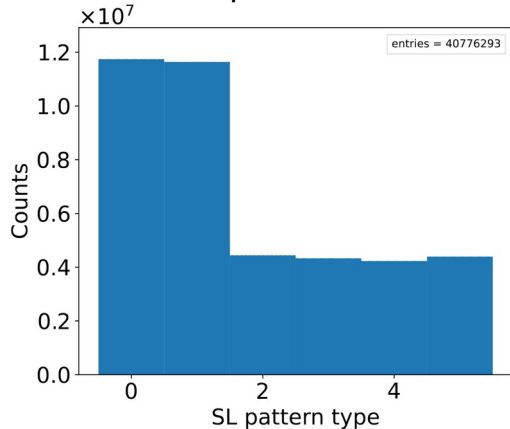
DT superlayer patterns

- Perform triggerless track reconstruction:
 - Treat all cell hits as part of potential track
- Superlayer pattern:
 - Require track-like shape of hits
 - Hit time difference limited by max. drift time:
$$\Delta t_{\text{max}} \leq t_{\text{drift,max}}$$
 - Only accept patterns with 4 cell hits

Superlayer pattern geometry:



Pattern occupancies:



- Cumulated pattern rate: **700 Hz / superlayer**

Atmospheric muon simulation: 450 Hz / superlayer

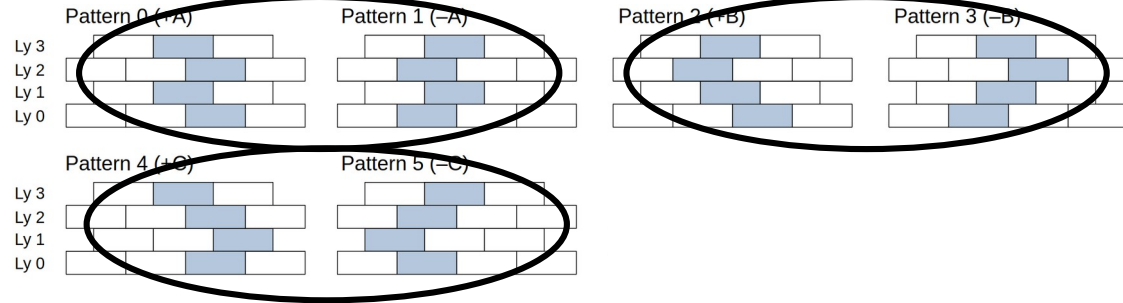
- Presence of noise-induced patterns in data

DT superlayer patterns

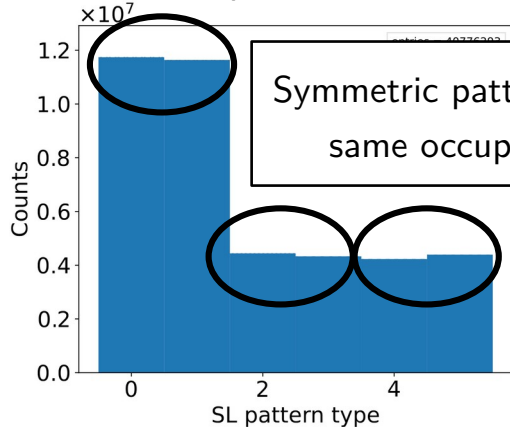
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Superlayer pattern geometry:



Pattern occupancies:



Simulated pattern rate: **700 Hz / superlayer**

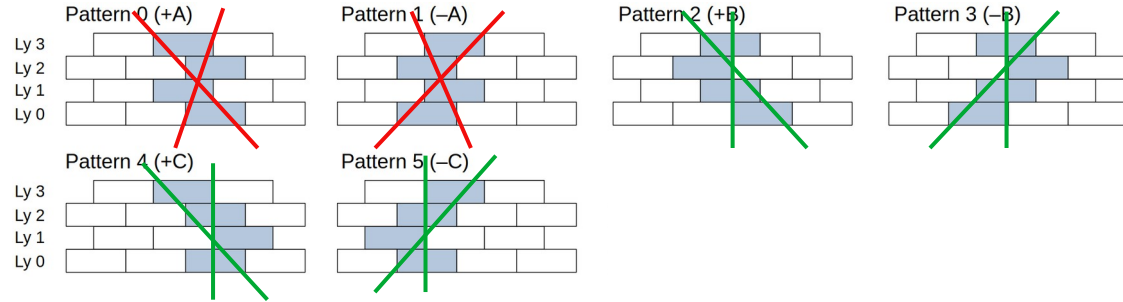
Atmospheric muon simulation: 450 Hz / superlayer

Presence of noise-induced patterns in data

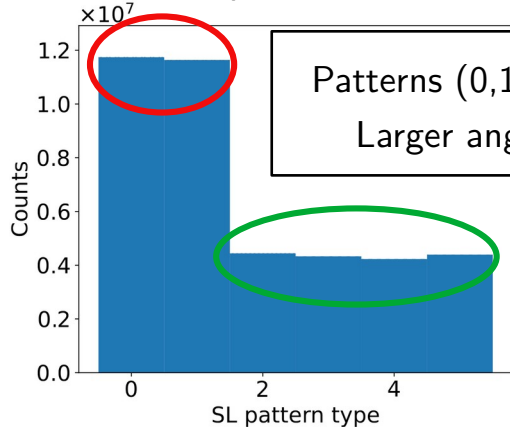
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 - Hit time difference limited by max. drift time:
$$\Delta t_{\max} \leq t_{\text{drift},\max}$$
 - Only accept patterns with 4 cell hits

Superlayer pattern geometry:



Pattern occupancies:



Patterns (0,1) have higher rate:
Larger angular acceptance

pattern rate: **700 Hz / superlayer**

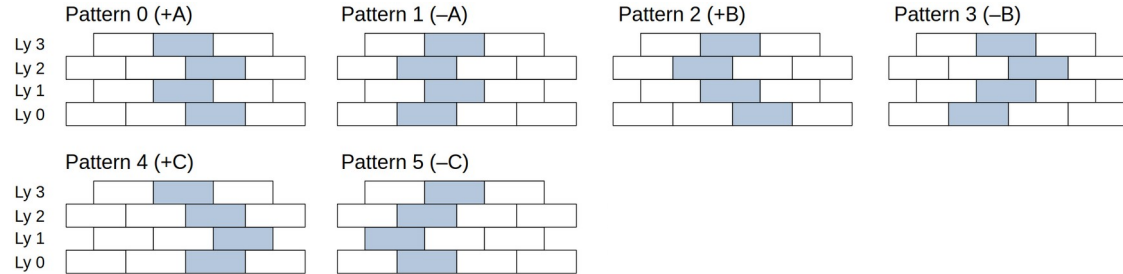
muon simulation: 450 Hz / superlayer

► Presence of noise-induced patterns in data

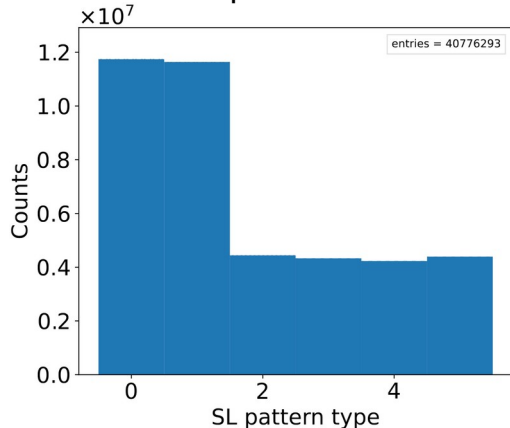
DT superlayer patterns

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 - Only accept patterns with 4 cell hits

Superlayer pattern geometry:



Pattern occupancies:



- ▶ Cumulated pattern rate: **700 Hz / superlayer**

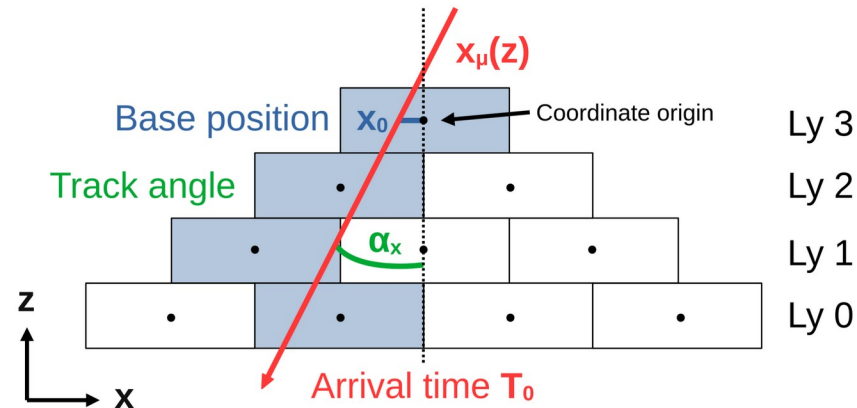
Atmospheric muon simulation: 450 Hz / superlayer

- ▶ Presence of noise-induced patterns in data

DT superlayer pattern fit

- Assumptions for reconstruction:
 - Constant drift velocity over full length: $\mathbf{v}_{\text{drift}} = \text{const.}$
 - Horizontal drift at height of wire: $\mathbf{z}_{\text{hit},ly} = \mathbf{z}_{\text{wire},ly}$
 - Straight muon tracks: $\mathbf{x}_{\text{hit},ly} = \mathbf{x}_0 + \mathbf{z}_{\text{wire},ly} \cdot \tan \alpha$
- Relation between cell hit position $\mathbf{x}_{\text{hit},ly}$ and timestamp $\mathbf{T}_{ly}$:
 - Arrival time $\mathbf{T}_0$ determines drift time: $\mathbf{t}_{\text{drift},ly} = \mathbf{T}_{ly} - \mathbf{T}_0$
 - Drift distance: $\mathbf{x}_{\text{drift},ly} = \mathbf{v}_{\text{drift}} \cdot \mathbf{t}_{\text{drift},ly}$
 - Left-right ambiguity of cell hit described by laterality $\mathbf{L}_{ly}$:

$$\mathbf{x}_{\text{hit},ly} = \mathbf{L}_{ly} \cdot \mathbf{x}_{\text{drift},ly} + \mathbf{x}_{\text{wire},ly}$$



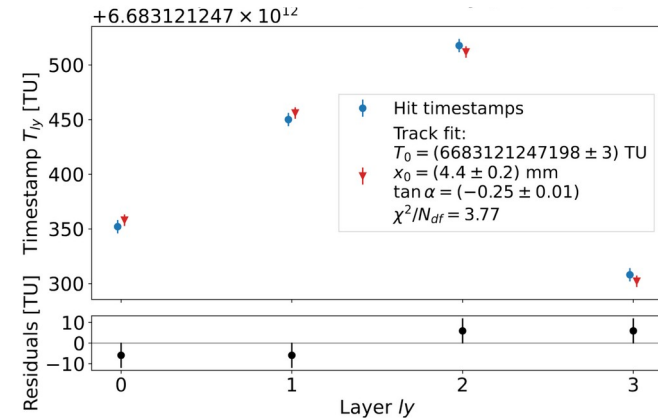
- Fit function of 4 timestamps ($\mathbf{T}_{ly}$) to 3 track parameters ($\mathbf{T}_0$, $\mathbf{x}_0$, $\tan \alpha$):

$$\mathbf{T}_{ly} = \mathbf{T}_0 + \mathbf{L}_{ly} / \mathbf{v}_{\text{drift}} \cdot (\mathbf{x}_0 + \mathbf{z}_{\text{wire},ly} \cdot \tan \alpha - \mathbf{x}_{\text{wire},ly})$$

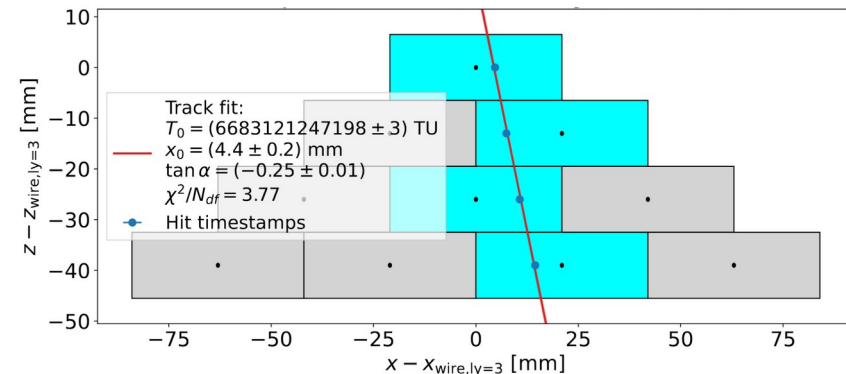
DT superlayer pattern fit

- Perform timestamp fits for all individual patterns
 - Consider all laterality combinations and keep only best fit
- Apply cuts to fit results to suppress noise hits:
 - Goodness-of-fit variable: $\chi^2/N_{df} < 10$
 - Unphysical drift times: $2 \text{ ns} \leq t_{\text{drift}} \leq t_{\text{drift,max}} - 2 \text{ ns} = 383 \text{ ns}$
- ▶ Fitted pattern rate: **400 Hz / superlayer**
 - ▶ Atmospheric muon simulation: 450 Hz / superlayer
 - ▶ Relatively good agreement

Example timestamp fit result:

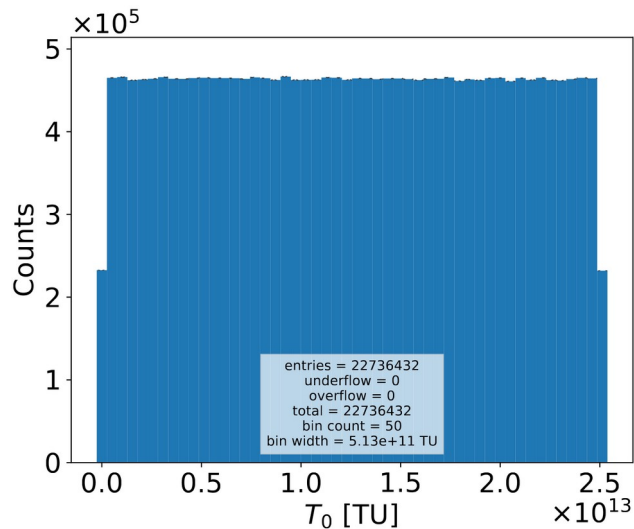


Reconstructed cell hit positions:

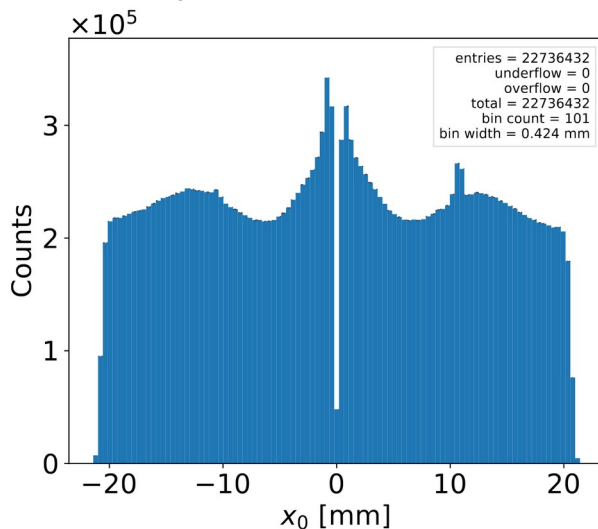


DT superlayer pattern fit results

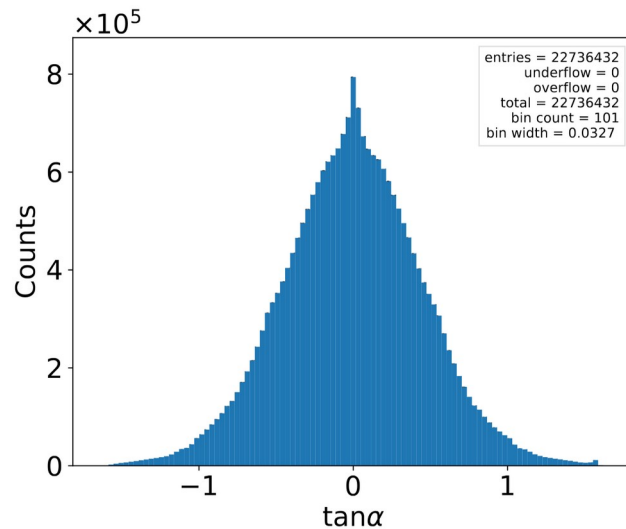
Arrival time T_0 :



Reference position in cell x_0 :

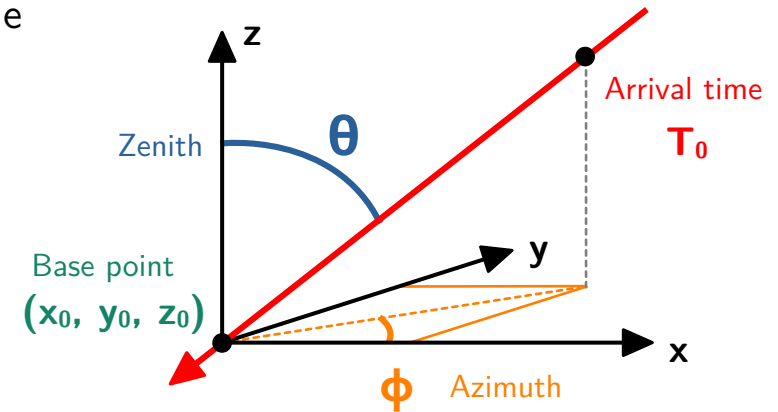
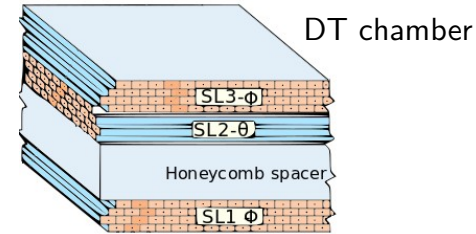


Track inclination $\tan \alpha$:



DT global track building

- Temporal matching of pattern fits in different superlayers
 - Select patterns close in time: $\Delta T_{0,SL} \leq 20 \text{ ns}$
 - Remove pattern ambiguities
 - Check consistency of SLs 1, 3 pattern fits
- Combination of superlayer projections into global coordinate frame
 - Global track parameters:
 - Arrival time T_0
 - Base point (x_0, y_0, z_0)
 - Azimuth angle Φ , zenith angle Θ
- ▶ Reconstructed global muon rate: **85 Hz**
- ▶ Global track parameters (mostly) behave as expected

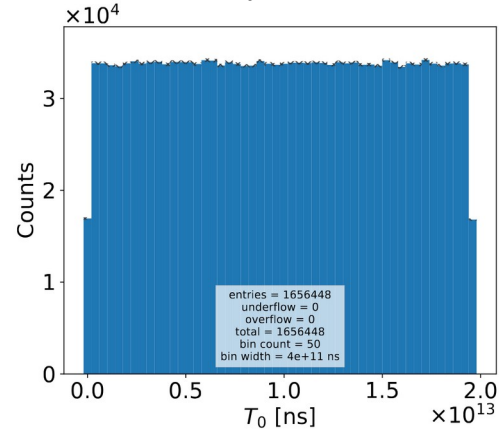


Atmospheric muon simulation: 196 Hz

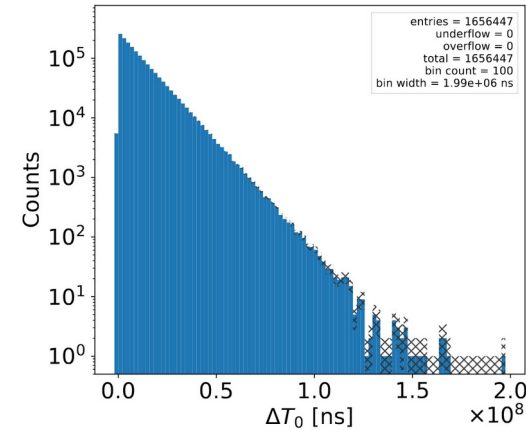
- ▶ Lower rate of reconstructed muons: Possibly because of assumptions in reconstruction

DT global track results

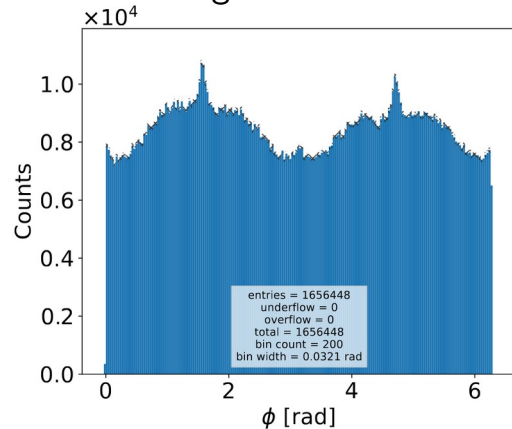
Arrival time T_0 :



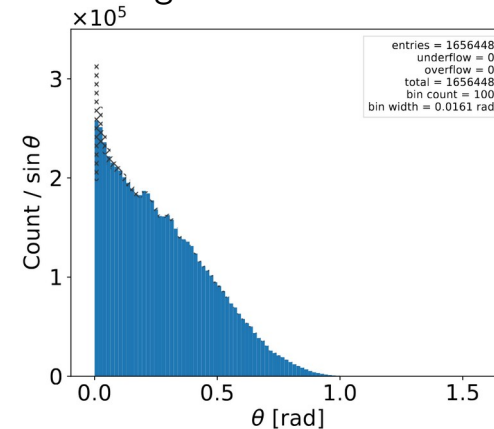
Time difference ΔT_0 of consecutive tracks:



Azimuth angle Φ :



Zenith angle Θ :

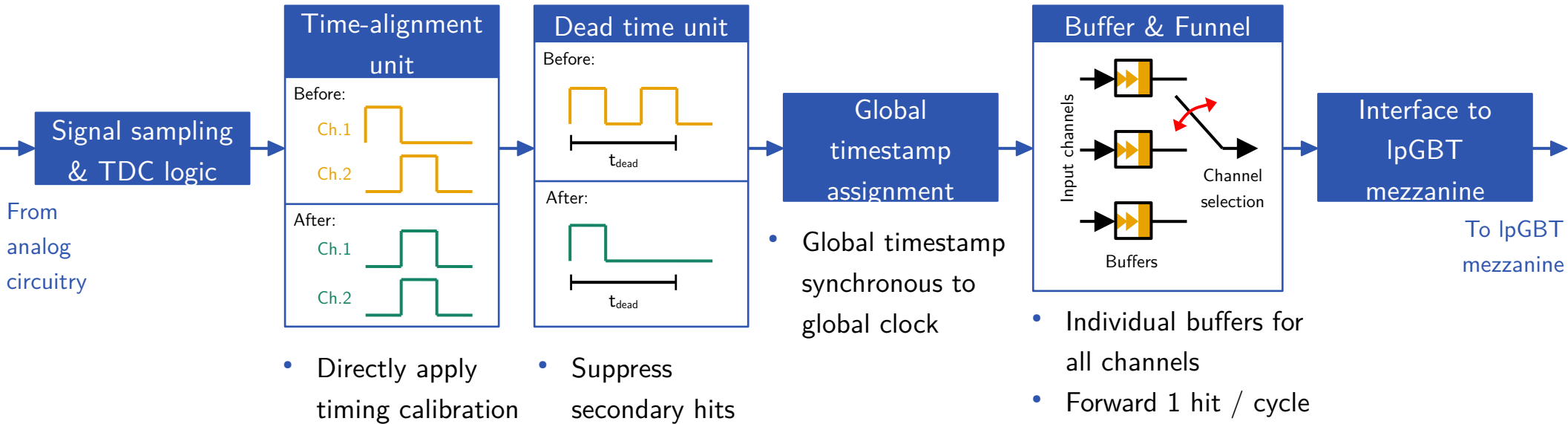


Backup (Pt. 3)

- SiPM-based scintillation detector
- Scintillator data analysis

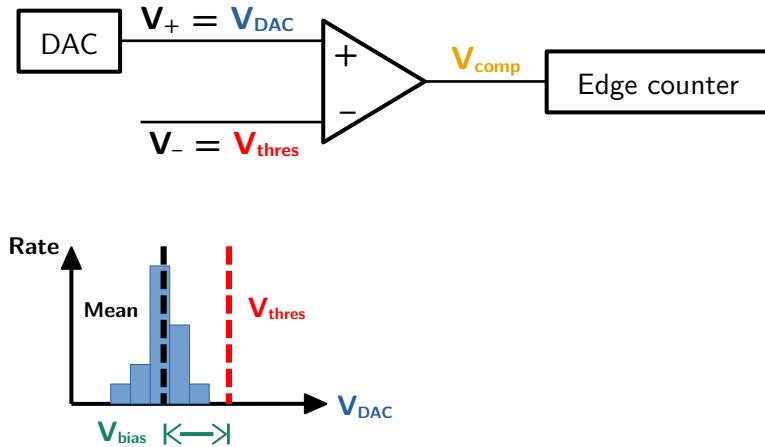
Scintillator readout electronics – FPGA firmware

- Completely self-developed readout firmware running on FPGA:

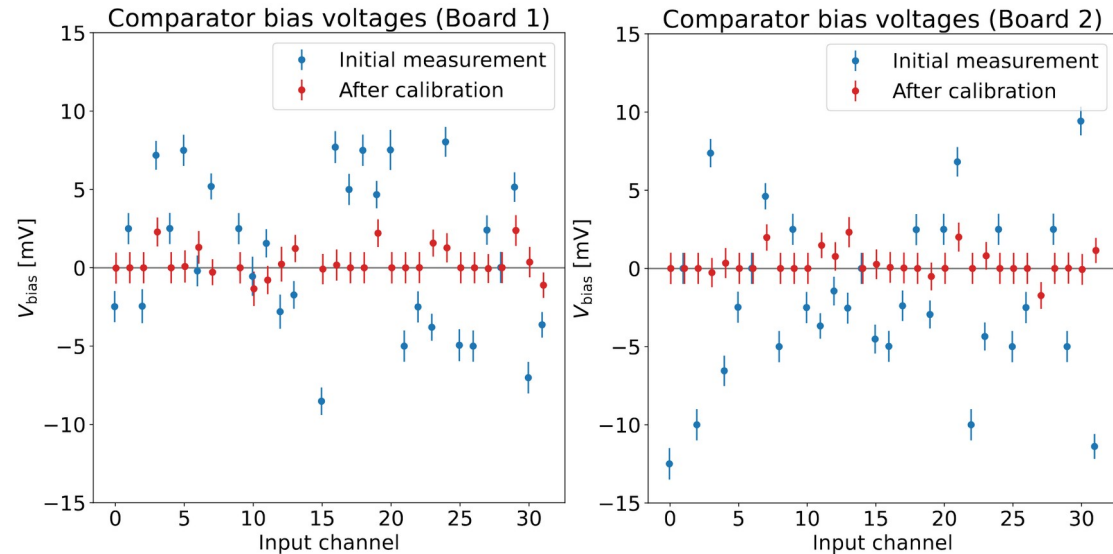


Input discriminator threshold calibration

- FPGA-internal comparators are non-ideal
 - Switching point shifted by bias voltage V_{bias}
- Bias measurement:
 - Scan comparator input voltage for fixed threshold
 - Determine comparator switching rate
 - Switching point lies at rate maximum

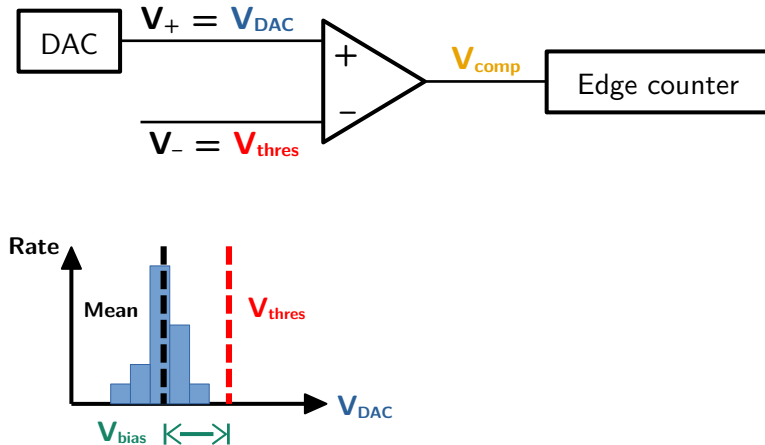


Measured bias before and after correction:

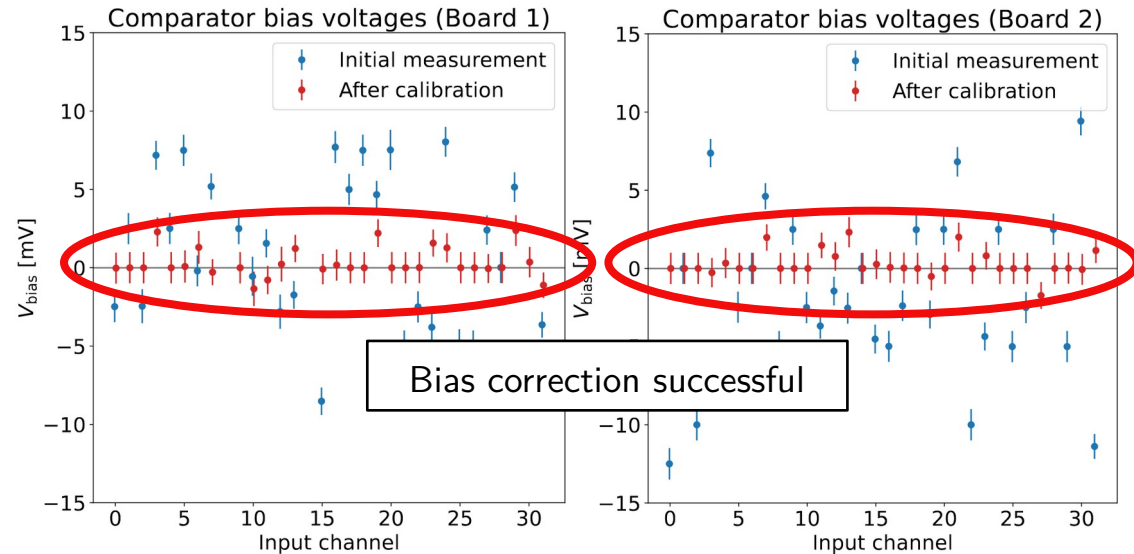


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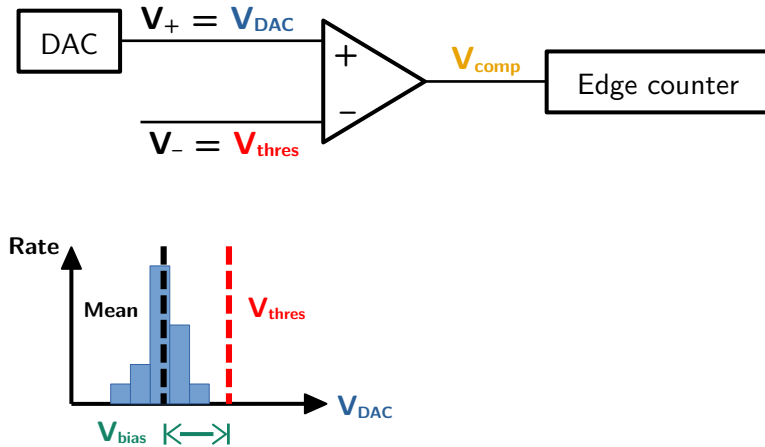


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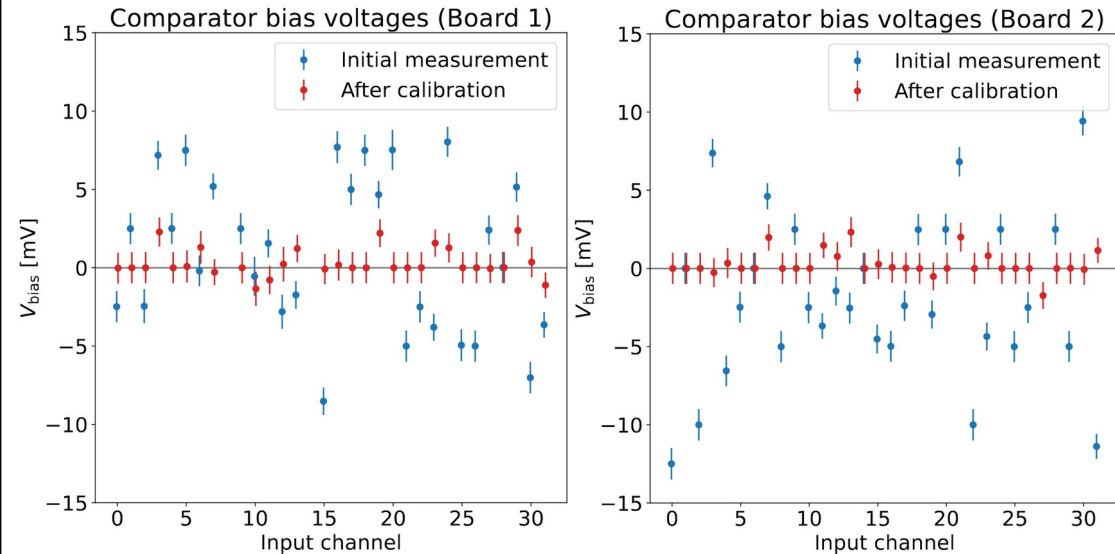


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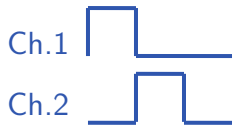
Measured bias before and after correction:



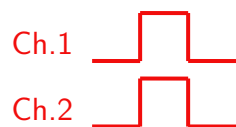
Input timing calibration with testpulses

- TDC logic implemented in FPGA has non-deterministic timing offset
 - Due to chip-internal calibration of deserializers on startup
 - Affects groups of inputs (I/O banks)
- Timing offset measurement:
 - Send simultaneous testpulses to all inputs
 - Measure response timestamps
- Time-alignment unit:
 - Programmable delay units in FPGA firmware
 - Calculate target delays from testpulse timing

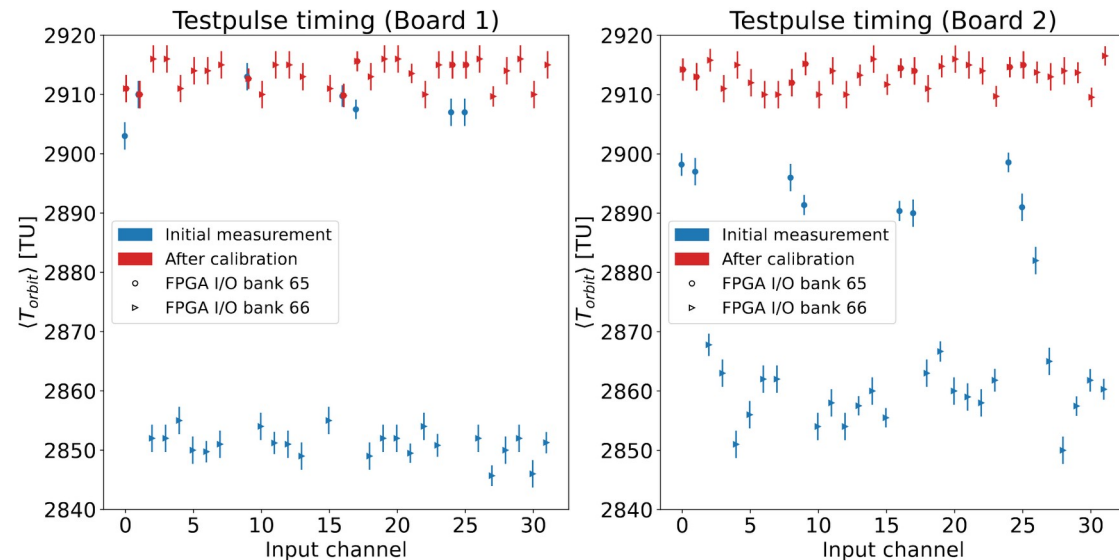
Before:



After:



- Measured testpulse response **before** and **after** programming delays:

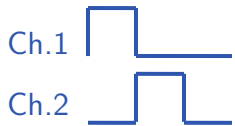


1 TU (timestamp unit) = 0.78 ns

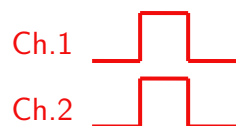
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- Time-alignment unit:
 - Programmable delay units in FPGA firmware
 - Calculate target delays from testpulse timing

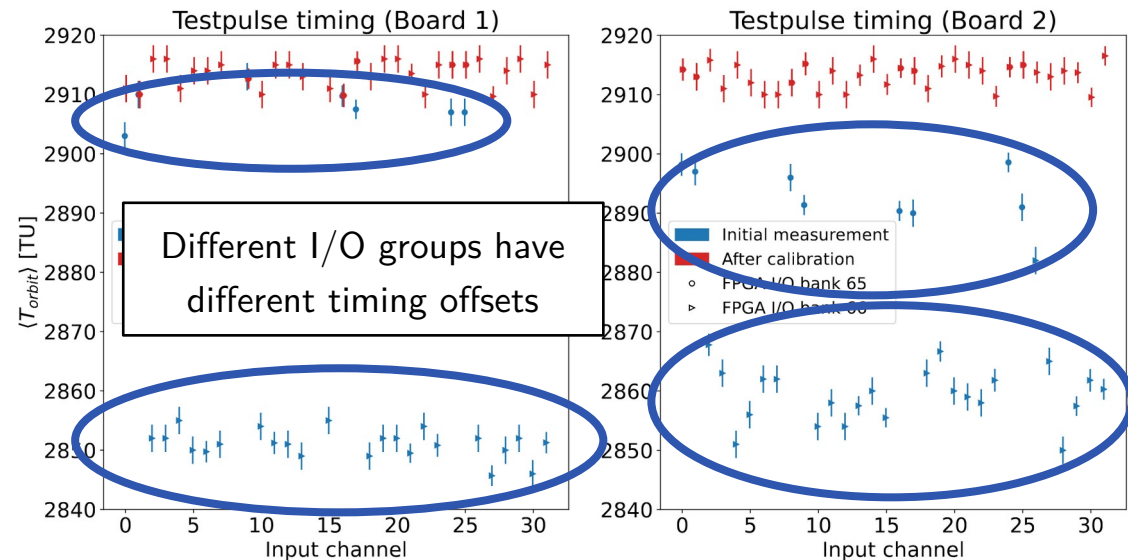
Before:



After:



- Measured testpulse response **before** and **after** programming delays:

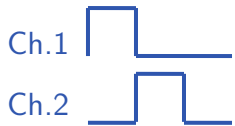


1 TU (timestamp unit) = 0.78 ns

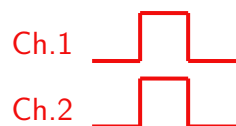
Input timing calibration with testpulses

- TDC logic implemented in FPGA has non-deterministic timing offset
 - Due to chip-internal calibration of deserializers on startup
 - Affects groups of inputs (I/O banks)
- Timing offset measurement:
 - Send simultaneous testpulses to all inputs
 - Measure response timestamps
- Time-alignment unit:
 - Programmable delay units in FPGA firmware
 - Calculate target delays from testpulse timing

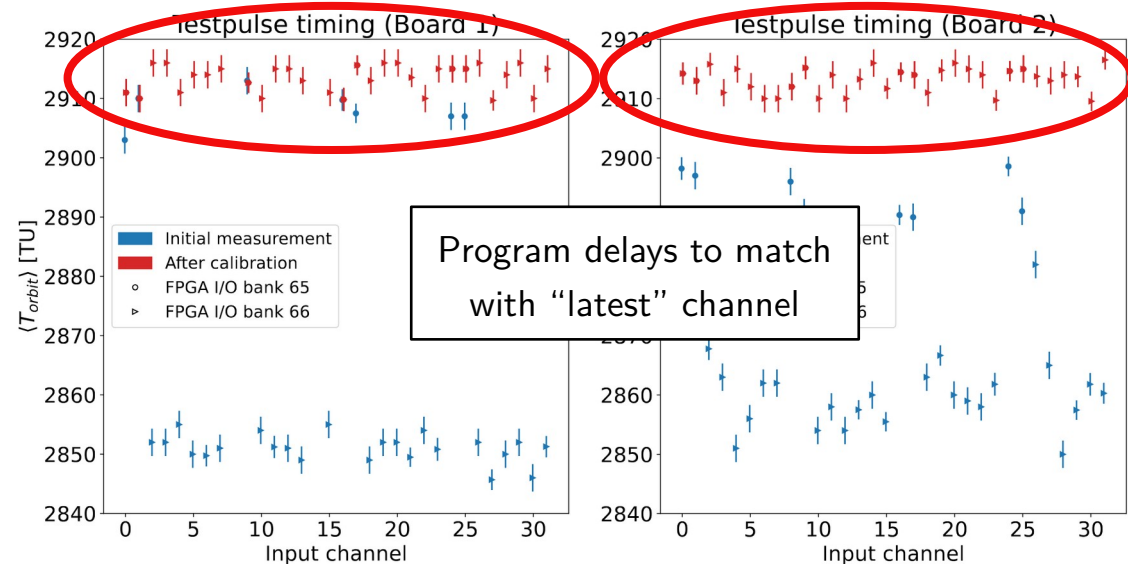
Before:



After:



- Measured testpulse response **before** and **after** programming delays:

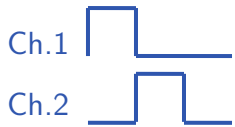


1 TU (timestamp unit) = 0.78 ns

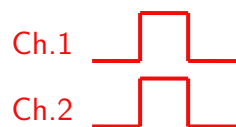
Input timing calibration with testpulses

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 - Send simultaneous testpulses to all inputs
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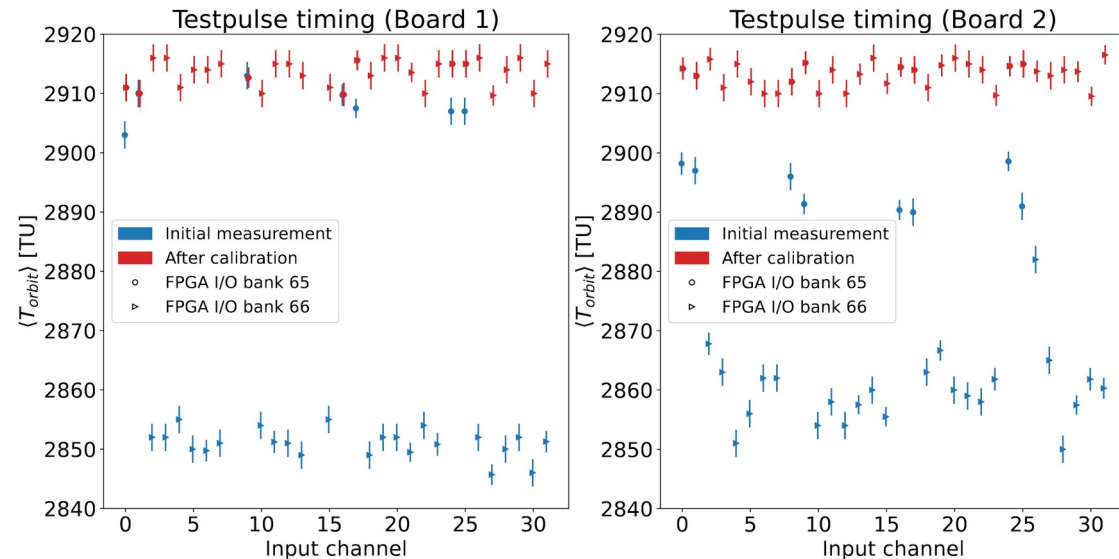
Before:



After:



- Measured testpulse response **before** and **after** programming delays:

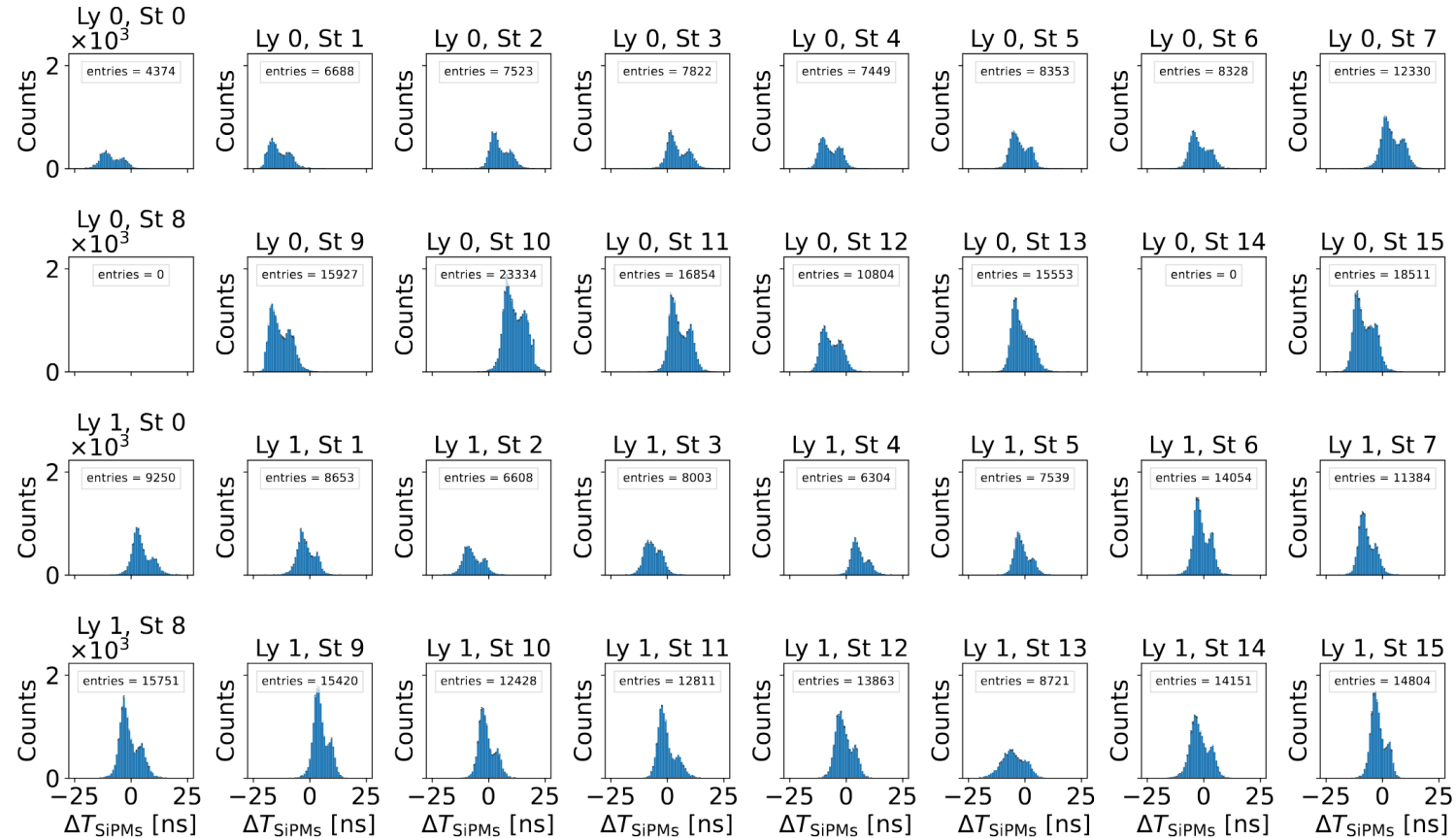


1 TU (timestamp unit) = 0.78 ns

Scintillator strip coincidences

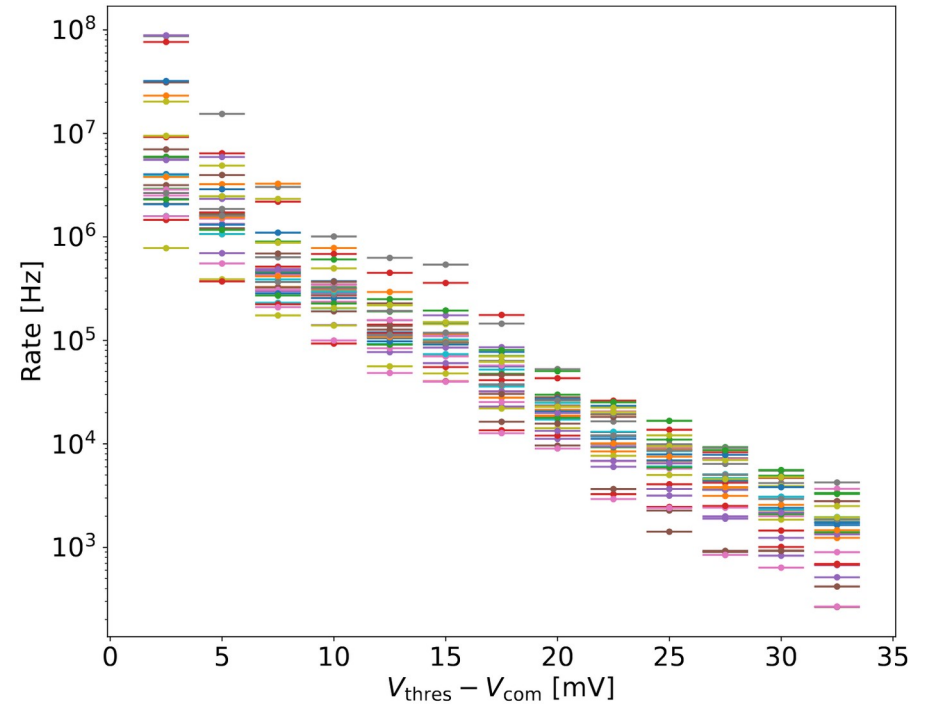
- Observe double-peak structure in time difference of SiPMs

Time difference ΔT_{SiPMs} of coincidences for all strips:



SiPM noise scan

- A



Backup (Pt. 4)

- Combined data analysis

Backup (Pt. 5)

- Simplified Monte Carlo simulation with cosmics